ESAT PRACTICE TEST SERIES

CHEMISTRY + BIOLOGY

LEVEL 2 · HARDER THAN ESAT

Chemistry + Biology

Mathematics 1 + Chemistry + Biology

Challenge material with denser interpretation, quantitative reasoning and longer solution chains.

MODULE
Mathematics 1

MODULE
Chemistry

MODULE
Biology

FULL SOLUTION BOOK

ThrivingScholars

81 preserved questions · three verified module keys

COMBINED ANSWER KEYS

Chemistry + Biology · Level 2 Harder than ESAT

This book compiles three existing 27-question modules into one 81-question pathway. No new questions have been generated and no source solution has been rewritten.

Composition: Mathematics 1 + Chemistry + Biology. Each module retains its original question order, answer and worked solution.

Mathematics 1

Engineering Mock Test 5 · preserved Mathematics 1 module

Q1 C	Q2 B	Q3 D	Q4 A	Q5 E
Q6 C	Q7 B	Q8 D	Q9 A	Q10 E
Q11 C	Q12 B	Q13 D	Q14 A	Q15 E
Q16 B	Q17 C	Q18 D	Q19 A	Q20 E
Q21 C	Q22 B	Q23 D	Q24 A	Q25 E
Q26 C	Q27 B			

Chemistry

Preserved Chemistry Harder than ESAT source set

Q1 A	Q2 D	Q3 F	Q4 H	Q5 C
Q6 G	Q7 B	Q8 G	Q9 E	Q10 F
Q11 C	Q12 B	Q13 E	Q14 A	Q15 C
Q16 D	Q17 C	Q18 B	Q19 E	Q20 B
Q21 A	Q22 D	Q23 F	Q24 D	Q25 F
Q26 E	Q27 A			

Biology

Preserved Biology Harder than ESAT source set

Q1 C	Q2 F	Q3 B	Q4 C	Q5 C
Q6 D	Q7 C	Q8 D	Q9 C	Q10 B
Q11 B	Q12 E	Q13 F	Q14 E	Q15 C
Q16 D	Q17 B	Q18 B	Q19 E	Q20 E
Q21 A	Q22 E	Q23 D	Q24 F	Q25 E
Q26 C	Q27 H			

LEVEL 2 · HARDER THAN ESAT

Mathematics 1

27 QUESTIONS · COMPLETE SOLUTIONS

Engineering Mock Test 5 · preserved Mathematics 1 module. Question wording, option order, answer and solution method are unchanged.

M1-01

Laws of indices and exact simplification

Answer C

QUESTION 1

Simplify fully

$$\frac{(50x^{3/2}y^{-1})^2(2x^{-1}y^{5/2})^3}{(20x^{1/2}y)^2},$$

where x and y are positive.

A $\frac{10y^3\sqrt{y}}{x}$

B $\frac{50y^3}{x}$

C $\frac{50y^3\sqrt{y}}{x}$

D $\frac{50\sqrt{y}}{x}$

E $\frac{250y^3\sqrt{y}}{x}$

WORKED SOLUTION

$(50x^{3/2}y^{-1})^2 = 2500x^3y^{-2}$ and $(2x^{-1}y^{5/2})^3 = 8x^{-3}y^{15/2}$.

So the numerator is $2500x^{3}y^{-2} \cdot 8x^{-3}y^{15/2} = 20000y^{11/2}$.

The denominator is $(20x^{1/2}y)^{2} = 400xy^{2}$.

Therefore

$$\frac{20000y^{11/2}}{400xy^{2}} = \frac{50y^{7/2}}{x} = \boxed{\frac{50y^{3}\sqrt{y}}{x}}.$$

M1-02

Successive percentage change and comparison

Answer B

QUESTION 2

The price of item P is reduced by 20%, and then a sales tax of 5% is applied to the reduced price.

The price of item Q is increased by 10%, and then reduced by 10%.

After these two-step changes, the final prices of P and Q are equal.

What was the ratio of the original price of P to the original price of Q?

- **A** $28 : 33$
- **B** $33 : 28$
- **C** $11 : 10$
- **D** $6 : 5$
- **E** $99 : 80$

WORKED SOLUTION

Final price of P is $0.8 \times 1.05P = 0.84P$.

Final price of Q is $1.1 \times 0.9Q = 0.99Q$.

So $0.84P = 0.99Q$, giving

$$\frac{P}{Q} = \frac{0.99}{0.84} = \frac{99}{84} = \frac{33}{28}.$$

M1-03

Algebraic fractions and hidden substitution

Answer D

QUESTION 3

For real x, define

$$E = \frac{\frac{1}{x-3} - \frac{1}{x+2}}{\frac{1}{x-3} + \frac{1}{x+2}}.$$

Given that $x > 3$ and

$$E + \frac{1}{E} = \frac{13}{2},$$

what is the value of x?

A $\frac{11}{2}$

B 8

C 12

D $\frac{31}{2}$

E 18

WORKED SOLUTION

Simplify first:

$$E = \frac{\frac{(x+2)-(x-3)}{(x-3)(x+2)}}{\frac{(x+2)+(x-3)}{(x-3)(x+2)}} = \frac{5}{2x-1}.$$

Let $t = E$. Then $t + \frac{1}{t} = \frac{13}{2}$, so

$$2t^2 - 13t + 2 = 0 = (2t-1)(t-6).$$

Hence $t = 6$ or $t = \frac{1}{6}$. Since $x > 3$, we have $0 < E < 1$, so $E = \frac{1}{6}$.

Therefore

$$\frac{5}{2x-1} = \frac{1}{6} \Rightarrow 2x - 1 = 30 \Rightarrow \boxed{x = \frac{31}{2}}.$$

M1-04

Inequalities with domain restrictions

Answer A

QUESTION 4

How many integer values of x satisfy all three conditions?

- $-5 \le x \le 5$
- $\frac{x^2+3x-10}{x^2-1}$ is defined
- $\frac{x^2+3x-10}{x^2-1} \le 1$

A 6

B 5

C 4

D 7

E 8

WORKED SOLUTION

Bring everything to one side:

$$\frac{x^2 + 3x - 10}{x^2 - 1} - 1 \leq 0 \Rightarrow \frac{3x - 9}{(x - 1)(x + 1)} \leq 0.$$

So we solve

$$\frac{x - 3}{(x - 1)(x + 1)} \leq 0, \qquad x \neq \pm 1.$$

A sign chart gives $(-\infty, -1) \cup (1, 3]$.

Intersecting with $-5 \leq x \leq 5$ gives $-5, -4, -3, -2, 2, 3$, so there are $\boxed{6}$ values.

M1-05

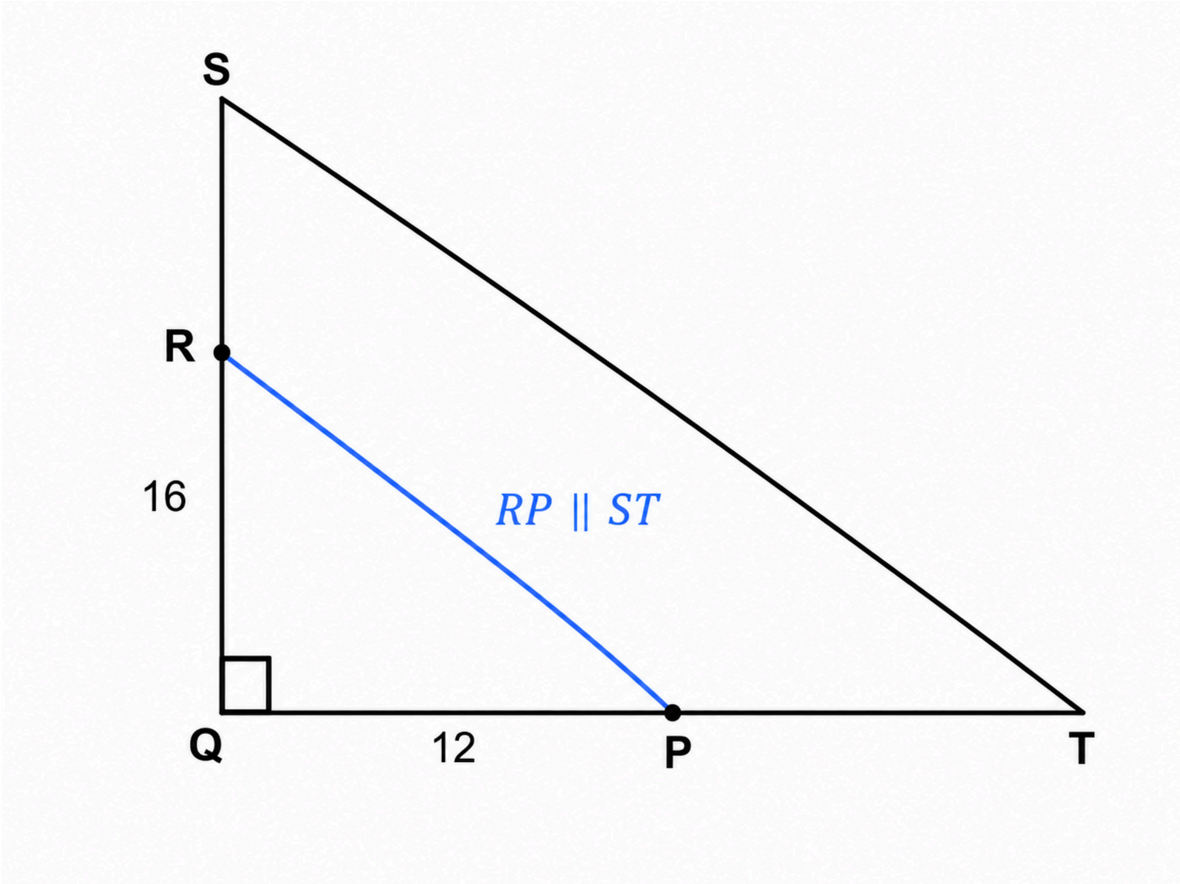
Similarity and area scaling in a right triangle

Answer **E**

QUESTION 5

Triangle SQT is right-angled at Q, with $SQ = 16$ cm and $QT = 12$ cm.

Point R lies on SQ such that $SR : RQ = 1 : 3$. Through R, a line parallel to ST meets QT at P.



What is the area of quadrilateral $SRPT$?

A $24\ \mathrm{cm}^2$

B $30\ \mathrm{cm}^2$

C $36\ \mathrm{cm}^2$

D $40\ \mathrm{cm}^2$

E $42\ \mathrm{cm}^2$

WORKED SOLUTION

The area of the whole triangle is $\frac{1}{2} \cdot 16 \cdot 12 = 96.$

Since $SR : RQ = 1 : 3$, we get $RQ = \frac{3}{4} \cdot 16 = 12.$

Because $RP \parallel ST$, triangles QRP and QST are similar with scale factor $\frac{QR}{QS} = \frac{12}{16} = \frac{3}{4}$.

So $QP = \frac{3}{4} \cdot 12 = 9$. Then area of triangle QRP is $\frac{1}{2} \cdot 12 \cdot 9 = 54$.

Hence quadrilateral $SRPT$ has area $96 - 54 = \boxed{42\ \text{cm}^2}$.

M1-06

Arithmetic sequences and common terms

Answer **C**

QUESTION 6

The nth term of sequence T is $4n - 3$.

The nth term of sequence V is $7n + p$, where p is an integer and $0 \le p < 7$.

The sequences have consecutive common terms 53 and 81.

What is the value of p?

- **A** 1
- **B** 2
- **C** 4
- **D** 5
- **E** 6

WORKED SOLUTION

Using the common term 53,

$$53 = 7n + p.$$

Since $53 \equiv 4 \pmod 7$, and $0 \leq p < 7$, we must have $\boxed{p = 4}$.

M1-07

Cylinders with swapped dimensions

Answer **B**

QUESTION 7

Two solid cylinders P and Q are defined using positive numbers x and y, where $x > y$.

- Cylinder P has diameter x and height y.
- Cylinder Q has diameter y and height x.

The total surface area of P is $\frac{8}{5}$ of the total surface area of Q.

The volume of P exceeds the volume of Q by $32\pi\ \text{cm}^3$.

What is the value of $x + y$?

A 10

B 12

C 14

D 16

E 18

WORKED SOLUTION

For a cylinder of diameter d and height h, total surface area is $2\pi(\frac{d}{2})^2 + 2\pi(\frac{d}{2})h$.

Thus

$$A_P = \pi\left(\frac{x^2}{2} + xy\right), \qquad A_Q = \pi\left(\frac{y^2}{2} + xy\right).$$

Set $x = ky$. Then

$$\frac{k^2 + 2k}{1 + 2k} = \frac{8}{5}.$$

So $5k^2 - 6k - 8 = 0$, giving the positive root $k = 2$. Hence $x = 2y$.

Now

$$V_P - V_Q = \frac{\pi}{4}(x^2y - y^2x) = \frac{\pi}{4}xy(x - y) = 32\pi.$$

Substitute $x = 2y$:

$$\frac{\pi}{4}(2y)(y)(y) = 32\pi \Rightarrow \frac{y^3}{2} = 32 \Rightarrow y^3 = 64 \Rightarrow y = 4.$$

Hence $x = 8$, so $x + y = \boxed{12}$.

M1-08

Ratio change after equal spending

Answer **D**

QUESTION 8

Pat and Alex have a combined total of £ 84. The ratio of Pat's money to Alex's money is $5 : 2$.

They each spend the same amount on sweets.

After this, Pat receives £ 6 and Alex receives £ 1. The ratio of their new amounts is then $3 : 1$.

How much did each of them spend?

A £ 3

B £ 4

C $£\ 4.25$

D $£\ 4.50$

E $£\ 6$

WORKED SOLUTION

Initially the amounts are 60 and 24.

If each spends s, then afterwards they have $60 - s$ and $24 - s$. After receiving the extra money, they have $66 - s$ and $25 - s$.

So

$$66 - s = 3(25 - s) \Rightarrow 66 - s = 75 - 3s \Rightarrow 2s = 9 \Rightarrow s = 4.5.$$

Each spent $\boxed{£\ 4.50}$.

M1-09 Coordinate geometry of a tangent Answer **A**

QUESTION 9

The circle $x^2 + y^2 = 25$ has centre at the origin.

A tangent to the circle passes through the point $(1, 7)$ and has positive gradient.

What is the equation of the tangent?

A $3x - 4y + 25 = 0$

B $4x - 3y + 25 = 0$

C $3x + 4y - 25 = 0$

D $4x + 3y - 25 = 0$

E $y = \frac{4}{3}x + \frac{17}{3}$

WORKED SOLUTION

Let the tangent be $y = mx + c$. Since it passes through $(1, 7)$, we have $c = 7 - m$.

The distance from the origin to the tangent must equal the radius 5:

$$\frac{|c|}{\sqrt{m^2 + 1}} = 5.$$

So

$$(7 - m)^2 = 25(m^2 + 1) \Rightarrow 12m^2 + 7m - 12 = 0 = (4m - 3)(3m + 4).$$

The positive gradient is $m = \frac{3}{4}$. Then $c = 7 - \frac{3}{4} = \frac{25}{4}$.

Hence

$$y = \frac{3}{4}x + \frac{25}{4} \Leftrightarrow \boxed{3x - 4y + 25 = 0}.$$

M1-10 Conditional probability with non-standard dice | Answer **E**

QUESTION 10

Two fair dice X and Y are available.

- The faces on X are labelled $1, 1, 2, 3, 4, 5$.
- The faces on Y are labelled $2, 3, 3, 4, 5, 6$.

One of the dice is chosen at random and rolled twice.

Given that the total shown is 7, what is the probability that the chosen die was X?

A $\frac{1}{3}$

B $\frac{3}{8}$

C $\frac{1}{2}$

D $\frac{3}{5}$

E $\frac{2}{5}$

WORKED SOLUTION

For die X, the ordered outcomes summing to 7 are $(2, 5), (5, 2), (3, 4), (4, 3)$: 4 outcomes out of 36. So $P(7 \mid X) = \frac{1}{9}$.

For die Y, 3 appears twice, so the ordered outcomes summing to 7 are 6 in total. Hence $P(7 \mid Y) = \frac{1}{6}$.

With equal prior probability of choosing each die,

$$P(X \mid 7) = \frac{\frac{1}{2} \cdot \frac{1}{9}}{\frac{1}{2} \cdot \frac{1}{9} + \frac{1}{2} \cdot \frac{1}{6}} = \frac{\frac{1}{9}}{\frac{1}{9} + \frac{1}{6}} = \frac{2}{5}.$$

M1-11 Exponential equation via substitution Answer C

QUESTION 11

The equation $2^x + 2^{2-x} = 5$ has two real solutions.

What is the larger possible value of x?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

Let $t = 2^x$. Then $2^{2-x} = \frac{4}{t}$, so

$$t + \frac{4}{t} = 5 \Rightarrow t^2 - 5t + 4 = 0 = (t-1)(t-4).$$

Thus $t = 1$ or $t = 4$, so $x = 0$ or $x = 2$. The larger value is $\boxed{2}$.

M1-12

Rearranging to make a variable the subject

Answer B

QUESTION 12

Suppose x and y satisfy

$$x = y + \frac{4}{y},$$

where $y > 0$ and $x > 4$.

Which of the following is the larger possible value of y in terms of x?

A $\dfrac{x - \sqrt{x^2 - 16}}{2}$

B $\dfrac{x + \sqrt{x^2 - 16}}{2}$

C $\dfrac{x + \sqrt{x^2 + 16}}{2}$

D $\dfrac{2}{x + \sqrt{x^2 - 16}}$

E $\dfrac{4}{x + \sqrt{x^2 - 16}}$

WORKED SOLUTION

Multiply by y:

$$xy = y^2 + 4 \Rightarrow y^2 - xy + 4 = 0.$$

Using the quadratic formula,

$$y = \frac{x \pm \sqrt{x^2 - 16}}{2}.$$

The larger possible value uses the positive sign, so

$$\boxed{y = \frac{x + \sqrt{x^2 - 16}}{2}}.$$

M1-13

Surface area comparison of a cylinder and a cube

Answer **D**

QUESTION 13

A solid cylinder has radius r cm and height h cm.

A cube has side length $\sqrt{\pi}\, r$ cm.

The total surface area of the cylinder is equal to four times the total surface area of the cube.

What is $\dfrac{h}{r}$?

A 7

B 9

C 10

D 11

E 12

WORKED SOLUTION

The cylinder’s total surface area is $2\pi r^2 + 2\pi rh = 2\pi r(r + h)$.

The cube’s total surface area is $6(\sqrt{\pi}r)^2 = 6\pi r^2$.

Given the cylinder equals four times the cube,

$$2\pi r(r + h) = 24\pi r^2.$$

Cancel $2\pi r$:

$$r + h = 12r \Rightarrow h = 11r.$$

Therefore $\frac{h}{r} = \boxed{11}$.

M1-14

Mean and median with an ordered list

Answer A

QUESTION 14

Consider the list

$$1,\ 2,\ x,\ x + 1,\ 2x - 1,\ 15,$$

where $2 < x < 8$.

The mean of the six numbers is one more than the median.

What is the value of x?

A $\frac{9}{2}$

B 4

C 5

D $\frac{11}{2}$

E 6

WORKED SOLUTION

Since $x > 2$, the list is already ordered. So the median is

$$\frac{x + (x + 1)}{2} = x + \frac{1}{2}.$$

The mean is one more than this, so the mean is $x + \frac{3}{2}$.

The mean is also

$$\frac{1 + 2 + x + (x + 1) + (2x - 1) + 15}{6} = \frac{4x + 18}{6}.$$

Therefore

$$\frac{4x + 18}{6} = x + \frac{3}{2} \Rightarrow 4x + 18 = 6x + 9 \Rightarrow 2x = 9 \Rightarrow \boxed{x = \frac{9}{2}}.$$

M1-15

Speed, time and average-rate reasoning

Answer E

QUESTION 15

A cyclist rides from A to B uphill at $12\ \mathrm{km\ h^{-1}}$, immediately turns round, and returns from B to A along the same route at $18\ \mathrm{km\ h^{-1}}$.

She spends exactly 6 minutes stationary at B.

The total time for the complete trip is 46 minutes.

What is the distance from A to B?

- **A** $3.6\ \mathrm{km}$
- **B** $4.0\ \mathrm{km}$
- **C** $4.2\ \mathrm{km}$
- **D** $4.5\ \mathrm{km}$
- **E** $4.8\ \mathrm{km}$

WORKED SOLUTION

Excluding the 6-minute stop, the travel time is 40 minutes = $\frac{2}{3}$ hours.

If the one-way distance is d km, then

$$\frac{\mathrm{d}}{12} + \frac{\mathrm{d}}{18} = \frac{2}{3}.$$

Since $\frac{1}{12} + \frac{1}{18} = \frac{5}{36}$,

$$\mathrm{d} \cdot \frac{5}{36} = \frac{2}{3} \Rightarrow \mathrm{d} = \frac{2}{3} \cdot \frac{36}{5} = \frac{24}{5} = 4.8.$$

So the distance is $\boxed{4.8 \text{ km}}$.

M1-16

Exact trigonometric values

Answer **B**

QUESTION 16

Given that

$$y = \frac{2\sin 15^\circ}{\cos 30^\circ},$$

what is the value of y^2?

A $\dfrac{4 - 2\sqrt{3}}{3}$

B $\dfrac{8 - 4\sqrt{3}}{3}$

C $\dfrac{8 + 4\sqrt{3}}{3}$

D $2 - \sqrt{3}$

E $4 - 2\sqrt{3}$

WORKED SOLUTION

Use

$$\sin 15^\circ = \sin(45^\circ - 30^\circ) = \frac{\sqrt{6} - \sqrt{2}}{4}, \qquad \cos 30^\circ = \frac{\sqrt{3}}{2}.$$

Hence

$$y = \frac{2 \cdot \frac{\sqrt{6}-\sqrt{2}}{4}}{\frac{\sqrt{3}}{2}} = \frac{\sqrt{6}-\sqrt{2}}{\sqrt{3}}.$$

Therefore

$$y^2 = \frac{(\sqrt{6}-\sqrt{2})^2}{3} = \frac{6+2-2\sqrt{12}}{3} = \frac{8-4\sqrt{3}}{3}.$$

M1-17

Inverse proportion in standard form

Answer C

QUESTION 17

The variable t is inversely proportional to the square of w.

When $w = 2 \times 10^3$, the value of t is 4.5×10^{-5}.

What is the value of w when $t = 8 \times 10^{-6}$?

- **A** 7.5×10^3
- **B** $3\sqrt{10} \times 10^2$
- **C** $1.5\sqrt{10} \times 10^3$
- **D** $2\sqrt{10} \times 10^3$
- **E** 4.5×10^3

WORKED SOLUTION

Since $t = \dfrac{k}{w^2}$, use the first condition:

$$k = (4.5 \times 10^{-5})(2 \times 10^3)^2 = (4.5 \times 10^{-5})(4 \times 10^6) = 180.$$

Now solve

$$8 \times 10^{-6} = \frac{180}{w^2} \Rightarrow w^2 = 2.25 \times 10^7.$$

So

$$w = \sqrt{2.25 \times 10^7} = 1.5\sqrt{10} \times 10^3.$$

M1-18

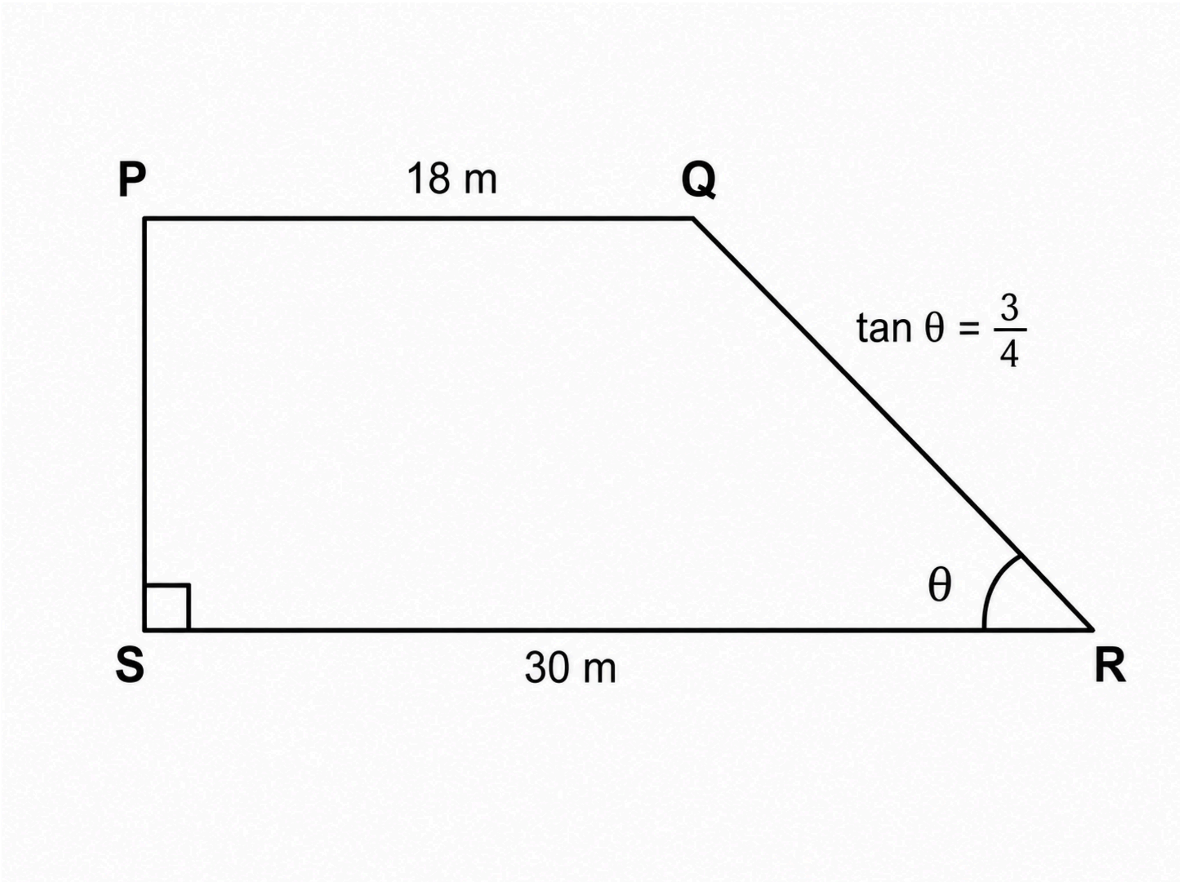
Trapezium area using trigonometry

Answer **D**

QUESTION 18

$PQRS$ is a trapezium in which $PQ \parallel SR$ and $PS \perp SR$.

Also, $PQ = 18$ m, $SR = 30$ m, and $\tan \angle QRS = \frac{3}{4}$.



What is the area of the trapezium?

A $180\ \text{m}^2$

B $192\ \mathrm{m}^2$

C $204\ \mathrm{m}^2$

D $216\ \mathrm{m}^2$

E $240\ \mathrm{m}^2$

WORKED SOLUTION

The difference between the parallel sides is $30 - 18 = 12$, which is the horizontal run of the sloping side.

If the height is h, then

$$\tan \angle QRS = \frac{h}{12} = \frac{3}{4} \Rightarrow h = 9.$$

So the area is

$$\frac{1}{2}(18 + 30)(9) = 24 \cdot 9 = \boxed{216\ \mathrm{m}^2}.$$

M1-19

Quadratic roots and discriminant reasoning

Answer A

QUESTION 19

The equation $x^2 + bx + 8 = 0$ has two real roots whose positive difference is 2. The sum of the roots is negative.

What is the value of b?

A 6

B 5

C 4

D -6

E -4

WORKED SOLUTION

If the roots are r and s, then $(r - s)^2 = (r + s)^2 - 4rs$.

Here $r + s = -b$ and $rs = 8$. Since the difference is 2,

$$4 = b^2 - 32 \Rightarrow b^2 = 36.$$

So $b = \pm 6$. The sum of the roots is negative, so $-b < 0 \Rightarrow b > 0$. Hence $\boxed{b = 6}$.

M1-20

Algebraic geometry with area change

Answer E

QUESTION 20

A rectangle has length $(2x - 1)$ cm and width $(x + 1)$ cm.

A larger rectangle is made by adding 3 cm to both the length and the width.

The larger rectangle has twice the area of the original rectangle.

What is the perimeter of the original rectangle?

A 24 cm

B 26 cm

C 28 cm

D 29 cm

E 30 cm

WORKED SOLUTION

Original area is $(2x - 1)(x + 1)$. New dimensions are $2x + 2$ and $x + 4$, so the new area is $(2x + 2)(x + 4)$.

Given new area is twice old area,

$$(2x + 2)(x + 4) = 2(2x - 1)(x + 1).$$

This simplifies to

$$x^2 - 4x - 5 = 0 = (x - 5)(x + 1).$$

So $x = 5$. The original dimensions are 9 and 6, so the perimeter is $2(9 + 6) = \boxed{30 \text{ cm}}$.

M1-21

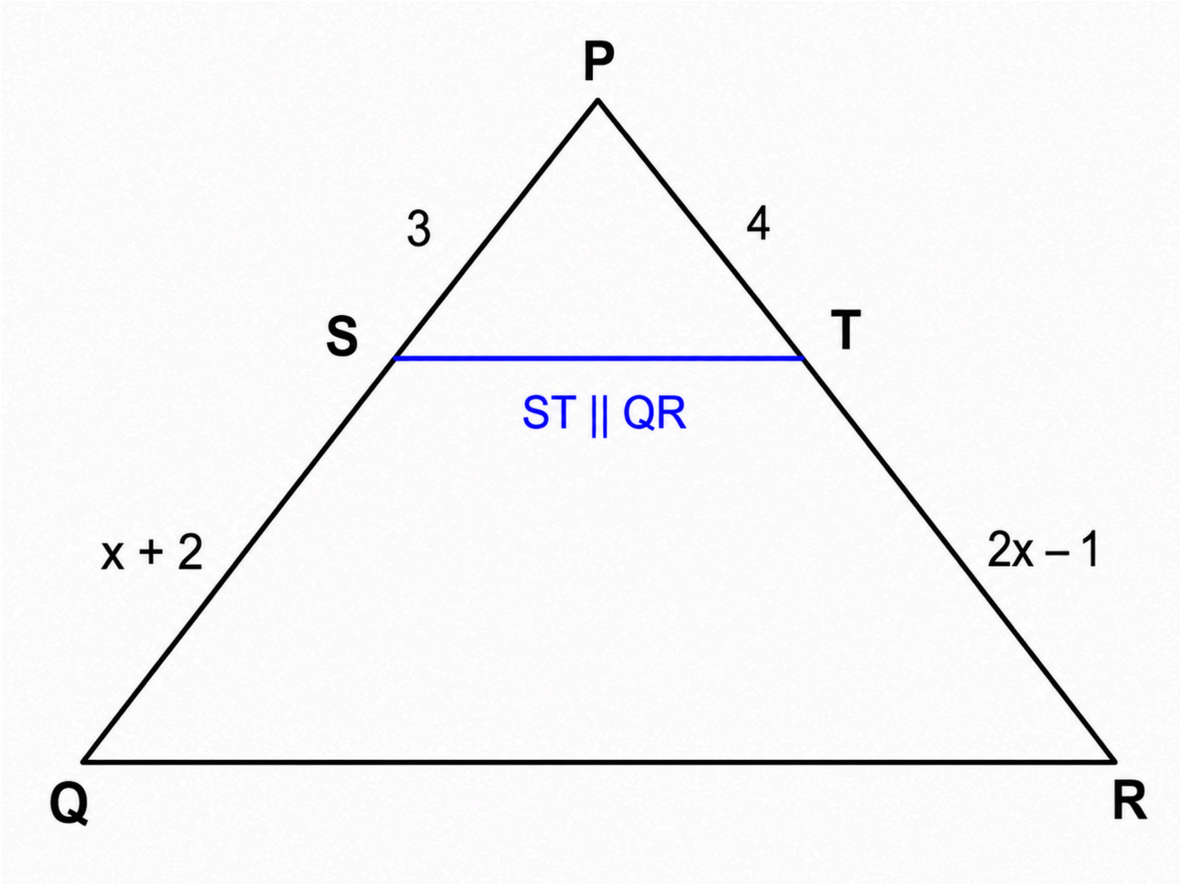
Similarity and area scaling

Answer **C**

QUESTION 21

In triangle PQR, point S lies on PQ and point T lies on PR. The line ST is parallel to QR.

The lengths are labelled as follows: $PS = 3$ cm, $SQ = (x + 2)$ cm, $PT = 4$ cm, $TR = (2x - 1)$ cm.



Given that the area of triangle PST is $12\ \mathrm{cm}^2$, what is the area of trapezium $STQR$?

A $96\ \mathrm{cm}^2$

B $120\ \mathrm{cm}^2$

C $135\ \mathrm{cm}^2$

D $144\ \mathrm{cm}^2$

E $147\ \mathrm{cm}^2$

WORKED SOLUTION

Since $ST \parallel QR$, triangles PST and PQR are similar, so

$$\frac{PS}{PQ} = \frac{PT}{PR}.$$

Now $PQ = 3 + (x + 2) = x + 5$ and $PR = 4 + (2x - 1) = 2x + 3$. Therefore

$$\frac{3}{x+5} = \frac{4}{2x+3}.$$

So $6x + 9 = 4x + 20$, giving $x = \frac{11}{2}$.

Then

$$\frac{PS}{PQ} = \frac{3}{21/2} = \frac{2}{7}.$$

Hence the area scale factor is $\left(\frac{2}{7}\right)^2 = \frac{4}{49}$.

So if area $PST = 12$, then area $PQR = 12 \cdot \frac{49}{4} = 147$.

Thus the area of trapezium $STQR$ is $147 - 12 = \boxed{135\ \text{cm}^2}$.

M1-22

Probability with two colours

Answer B

QUESTION 22

Charlie has a bowl containing red sweets and green sweets only. There are 9 more red sweets than green sweets.

Two sweets are chosen at random without replacement.

The probability that the two sweets are the same colour is $\frac{4}{7}$.

How many sweets are there in the bowl altogether?

A 18

B 21

C 24

D 27

E 30

WORKED SOLUTION

Let the number of green sweets be g. Then the number of red sweets is $g+9$, so total sweets are $2g+9$.

The probability condition is

$$\frac{\binom{g+9}{2}+\binom{g}{2}}{\binom{2g+9}{2}}=\frac{4}{7}.$$

This simplifies to

$$(g-6)(g+18)=0.$$

So $g=6$. Then red sweets $=15$, giving total sweets $=21$.

M1-23

Successive translations and vector equations

Answer **D**

QUESTION 23

The point $(-1,5)$ is translated by two successive vectors.

The first translation is by $\begin{pmatrix} 3p \\ 2-q \end{pmatrix}$, and the second is by $\begin{pmatrix} q+1 \\ p-3 \end{pmatrix}$.

The final image is $(8,4)$.

What is the value of p^2+q^2?

A 4

B 5

C 6

D 8

E 10

WORKED SOLUTION

The total translation vector is

$$\begin{pmatrix} 3p + q + 1 \\ p - q - 1 \end{pmatrix}.$$

Since $(-1, 5) \to (8, 4)$, the net translation is $\begin{pmatrix} 9 \\ -1 \end{pmatrix}$.

So

$$3p + q = 8, \qquad p - q = 0.$$

Thus $p = q$, so $4p = 8 \Rightarrow p = 2$, hence $q = 2$.

Therefore $p^2 + q^2 = 4 + 4 = \boxed{8}$.

M1-24 Quadratic graphs and minimum values — Answer A

QUESTION 24

Consider the family of graphs

$$y = ax^2 + 2(a - 1)x + a + 3.$$

What is the complete range of values of a for which the minimum point of the graph lies above the x-axis?

A $a > \frac{1}{5}$

B $a > 0$

C $a \geq \frac{1}{5}$

D $a < 0$

E $0 < a < \frac{1}{5}$

WORKED SOLUTION

For there to be a minimum, we need $a > 0$.

For $y = ax^2 + bx + c$, the minimum value is $c - \frac{b^2}{4a}$.

Here $b = 2(a-1)$ and $c = a + 3$, so the minimum value is

$$a + 3 - \frac{(2(a-1))^2}{4a} = a + 3 - \frac{(a-1)^2}{a} = 5 - \frac{1}{a}.$$

We need this to be positive:

$$5 - \frac{1}{a} > 0.$$

Since $a > 0$, this gives $5a > 1 \Rightarrow a > \frac{1}{5}$.

M1-25 Counting constrained selections Answer **E**

QUESTION 25

An online company sells storage containers. The following items are available:

Capacity of container	Number available
2 litres	2
3 litres	3

Capacity of container	Number available
5 litres	4

A customer chooses a selection of containers with total capacity 23 litres.

Different selections are distinguished only by how many containers of each size are chosen.

If all possible selections are equally likely, what is the probability that the chosen selection contains at least one 2-litre container?

A $\frac{1}{3}$

B $\frac{1}{2}$

C $\frac{3}{5}$

D $\frac{3}{4}$

E $\frac{2}{3}$

WORKED SOLUTION

Let a, b, c be the numbers of 2-, 3- and 5-litre containers. Then

$$2a + 3b + 5c = 23,$$

with $0 \le a \le 2, 0 \le b \le 3, 0 \le c \le 4.$

The valid selections are

- $(0, 1, 4)$
- $(1, 2, 3)$
- $(2, 3, 2)$

Two of the three contain at least one 2-litre container, so the probability is $\boxed{\frac{2}{3}}$.

M1-26

Similarity and three-dimensional scaling

Answer C

QUESTION 26

The space diagonal of a cuboid is $7\sqrt{13}$ cm.

A similar cuboid has volume 8 times the volume of the original cuboid.

What is the space diagonal of the larger cuboid?

A $7\sqrt{26}$ cm

B $14\sqrt{7}$ cm

C $14\sqrt{13}$ cm

D $21\sqrt{13}$ cm

E $28\sqrt{13}$ cm

WORKED SOLUTION

A volume scale factor of 8 means a linear scale factor of $\sqrt[3]{8} = 2$.

So the new diagonal is

$$2 \cdot 7\sqrt{13} = \boxed{14\sqrt{13}\text{ cm}}.$$

M1-27

Relative speed in a race

Answer B

QUESTION 27

Alex, Cameron and Sam are running a $400\ \mathrm{m}$ race. They each run at a different constant speed.

When Alex finishes, Cameron has run $300\ \mathrm{m}$ and Sam has run $240\ \mathrm{m}$.

Alex beats Sam by $33\frac{1}{3}$ seconds.

What is Cameron's speed?

A $5\ \mathrm{m\ s^{-1}}$

B $6\ \mathrm{m\ s^{-1}}$

C $6.4\ \mathrm{m\ s^{-1}}$

D $7\ \mathrm{m\ s^{-1}}$

E $8\ \mathrm{m\ s^{-1}}$

WORKED SOLUTION

Let Alex's speed be A, Cameron's speed be C, and Sam's speed be S.

When Alex finishes, the elapsed time is $\frac{400}{\mathrm{A}}$. In that time, Cameron runs 300 m, so

$$\mathrm{C} \cdot \frac{400}{\mathrm{A}} = 300 \Rightarrow \frac{\mathrm{C}}{\mathrm{A}} = \frac{3}{4}.$$

Similarly, Sam runs 240 m, so

$$\mathrm{S} \cdot \frac{400}{\mathrm{A}} = 240 \Rightarrow \frac{\mathrm{S}}{\mathrm{A}} = \frac{3}{5}.$$

Thus $S = \frac{3}{5}A$.

Alex beats Sam by $\frac{100}{3}$ s, so

$$\frac{400}{S} - \frac{400}{A} = \frac{100}{3}.$$

Substitute $S = \frac{3}{5}A$:

$$\frac{400}{3A/5} - \frac{400}{A} = \frac{100}{3} \Rightarrow \frac{2000}{3A} - \frac{400}{A} = \frac{100}{3} \Rightarrow \frac{800}{3A} = \frac{100}{3}.$$

So $A = 8\ \mathrm{m\ s^{-1}}$. Hence

$$C = \frac{3}{4}A = 6\ \mathrm{m\ s^{-1}}.$$

Therefore Cameron’s speed is $\boxed{6\ \mathrm{m\ s^{-1}}}$.

LEVEL 2 · HARDER THAN ESAT

Chemistry

27 QUESTIONS · COMPLETE SOLUTIONS

Preserved Chemistry Harder than ESAT source set. Question wording, option order, answer and solution method are unchanged.

C-01

Atomic structure and isotopes

Answer A

QUESTION 1

An ion X^- contains 18 electrons and 18 neutrons.

A sample of element X contains only two isotopes: the isotope described above and another isotope whose mass number is 2 greater.

The relative atomic mass of X in this sample is 35.6.

What percentage of naturally occurring X atoms are the heavier isotope?

A 30%

B 20%

C 25%

D 35%

E 40%

F 60%

WORKED SOLUTION

Because X^- has 18 electrons, neutral X has 17 electrons and therefore 17 protons.

The stated isotope therefore has mass number

$$17 + 18 = 35.$$

The two isotopes in the sample are 35 and 37. Let the fraction of the heavier isotope be p. Then

$$35(1 - p) + 37p = 35.6.$$

So

$$35 + 2p = 35.6 \Rightarrow p = 0.30.$$

The heavier isotope has abundance $\boxed{30\%}$, so the answer is **A**.

C-02

Precipitation and dissolved ions

Answer **D**

QUESTION 2

$25.0\ \mathrm{cm}^3$ of $0.120\ \mathrm{mol\ dm}^{-3}$ aluminium sulfate solution is mixed with $10.0\ \mathrm{cm}^3$ of $0.300\ \mathrm{mol\ dm}^{-3}$ barium chloride solution.

Barium sulfate is insoluble. Assume all other substances remain completely dissociated in solution.

What is the **total amount, in moles, of dissolved ions remaining after precipitation is complete**?

- **A** 0.006
- **B** 0.009
- **C** 0.012
- **D** 0.018
- **E** 0.015

F 0.021

WORKED SOLUTION

Moles of $Al_2(SO_4)_3$:

$$0.120 \times 0.0250 = 0.00300.$$

This gives

$$0.00600 \text{ mol } Al^{3+}, \qquad 0.00900 \text{ mol } SO_4^{2-}.$$

Moles of $BaCl_2$:

$$0.300 \times 0.0100 = 0.00300.$$

This gives 0.00300 mol Ba^{2+} and 0.00600 mol Cl^-.

The precipitation is

$$Ba^{2+} + SO_4^{2-} \rightarrow BaSO_4(s).$$

All Ba^{2+} is removed, leaving

$$0.00900 - 0.00300 = 0.00600 \text{ mol } SO_4^{2-}.$$

Total dissolved ions remaining:

$$0.00600 + 0.00600 + 0.00600 = \boxed{0.0180 \text{ mol}}.$$

The answer is **D**.

C-03

Structure and bonding

Answer **F**

QUESTION 3

The properties of five substances are shown.

substance	melting point	conducts when solid	conducts when molten	soluble in water
P	high	no	yes	yes
Q	very high	yes	yes	no
R	very high	no	no	no
S	moderate	yes	yes	no
T	very low	no	no	no

Which row gives the most likely structural types of **P, Q and R**, respectively?

A ionic, metallic, simple molecular

B giant covalent, metallic, ionic

C ionic, ionic, giant covalent

D metallic, graphite-like giant covalent, ionic

E simple molecular, metallic, giant covalent

F ionic, metallic, giant covalent

WORKED SOLUTION

P has the characteristic pattern of an ionic substance: high melting point, no conduction as a solid, conduction when molten and possible water solubility.

Q has a very high melting point, conducts in both the solid and molten states and is insoluble in water, consistent with a metallic structure.

R has a very high melting point and no mobile charge carriers, consistent with a giant covalent structure such as silicon dioxide.

Therefore the answer is **F**.

C-04

Visible chemical changes

Answer H

QUESTION 4

The following pairs of aqueous substances are mixed separately.

1. silver nitrate and potassium bromide
2. chlorine and potassium iodide
3. hydrochloric acid and sodium hydroxide, with no indicator present
4. sodium nitrate and potassium chloride

Which mixtures produce a **visible chemical change**?

- **A** 1 only
- **B** 2 only
- **C** 1 and 3 only
- **D** 2 and 3 only
- **E** 1, 2 and 3 only
- **F** 1, 2 and 4 only
- **G** all four
- **H** 1 and 2 only

WORKED SOLUTION

Mixture 1 forms a precipitate of silver bromide, so there is a visible change.

Mixture 2 undergoes halogen displacement:

$$Cl_2 + 2I^- \rightarrow 2Cl^- + I_2,$$

giving a colour change.

Mixture 3 undergoes neutralisation, but without an indicator there is no necessary visible change. Mixture 4 leaves all ions in solution.

Therefore **1 and 2 only** are visible, answer **H**.

C-05

Reactivity and extraction

Answer **C**

QUESTION 5

Metal M has the following properties.

- It does not react rapidly with cold water.
- It reacts with steam.
- Its oxide can be reduced by carbon.
- It displaces copper from aqueous copper(II) sulfate.

Which could be M?

A aluminium

B calcium

C iron

D copper

E magnesium

F potassium

WORKED SOLUTION

The metal must be reactive enough to react with steam and displace copper, but it must lie below carbon because its oxide can be reduced by carbon.

Iron satisfies all of these conditions. Aluminium and magnesium are too reactive for their oxides to be reduced by carbon under the standard extraction model;

calcium and potassium are far more reactive; copper cannot displace itself.

The answer is **C: iron**.

C-06

Oxidising agents

Answer G

QUESTION 6

In which reactions is the **first reactant written** acting as an oxidising agent?

1. $$Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$$
2. $$Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$$
3. $$MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$$
4. $$2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$$

A 1 only

B 1 and 3 only

C 1 and 4 only

D 2 and 3 only

E 2, 3 and 4 only

F all four

G 1, 3 and 4 only

WORKED SOLUTION

An oxidising agent is itself reduced.

1. Chlorine changes from oxidation state 0 to −1, so Cl_2 is reduced and is an oxidising agent.

2. Zinc changes from 0 to +2, so it is oxidised and acts as a reducing agent.

3. Manganese changes from +4 in MnO_2 to +2 in $MnCl_2$, so MnO_2 is reduced.

4. $Fe^{3+} \rightarrow Fe^{2+}$ is reduction.

Thus the first reactant is an oxidising agent in **1, 3 and 4**, answer **G**.

C-07

Balancing an unfamiliar redox reaction

Answer **B**

QUESTION 7

A divalent metal M reacts with nitric acid to form:

- $M(NO_3)_2$,
- water,
- a single oxide of nitrogen X.

Exactly 0.075 mol of M reacts with 0.300 mol of nitric acid. No other products are formed.

What is the empirical formula of X?

A N_2O

B NO_2

C NO

D N_2O_3

E N_2O_5

WORKED SOLUTION

0.075 mol of M forms 0.075 mol of $M(NO_3)_2$, using

$$2(0.075) = 0.150 \text{ mol N atoms}$$

in nitrate.

Total nitrogen supplied is 0.300 mol, so nitrogen in X is

$$0.300 - 0.150 = 0.150 \text{ mol N atoms.}$$

The 0.300 mol of nitric acid supplies 0.300 mol of H atoms, so 0.150 mol of water is formed.

Total oxygen supplied is

$$3(0.300) = 0.900 \text{ mol O atoms.}$$

Oxygen used in nitrate is $6(0.075) = 0.450$ mol, and oxygen used in water is 0.150 mol. Therefore oxygen in X is

$$0.900 - 0.450 - 0.150 = 0.300.$$

Thus

$$N : O = 0.150 : 0.300 = 1 : 2,$$

so $X = \boxed{NO_2}$. The answer is **B**.

C-08

Acid-base mixture and pH

Answer G

QUESTION 8

50.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ hydrochloric acid is mixed with 150 cm^3 of $0.0125 \text{ mol dm}^{-3}$ calcium hydroxide solution.

Which range contains the pH of the final solution?

A $0 < \text{pH} < 1$

B $1 < \text{pH} < 2$

C $3 < \text{pH} < 7$

D $\text{pH} = 7$

E $7 < \text{pH} < 12$

F $12 < \text{pH} < 14$

G $2 < \text{pH} < 3$

WORKED SOLUTION

Moles of H^+:

$$0.100(0.0500) = 0.00500.$$

Moles of $Ca(OH)_2$:

$$0.0125(0.150) = 0.001875.$$

This supplies

$$2(0.001875) = 0.00375 \text{ mol } OH^-.$$

Hydrogen ions are in excess:

$$0.00500 - 0.00375 = 0.00125 \text{ mol } H^+.$$

Total volume is 0.200 dm^3, so

$$[H^+] = \frac{0.00125}{0.200} = 0.00625 \text{ mol dm}^{-3}.$$

This concentration lies between 10^{-3} and 10^{-2} mol dm^{-3}. Therefore the pH lies between 2 and 3:

$$\boxed{2 < \text{pH} < 3}.$$

The answer is **G**.

C-09

Deducing properties of an acid

Answer E

QUESTION 9

An acid solution has

$$\text{concentration} = 0.300\ \text{mol dm}^{-3}$$

and

$$\text{mass concentration} = 29.4\ \text{g dm}^{-3}.$$

A $25.0\ \text{cm}^3$ sample is completely neutralised by $45.0\ \text{cm}^3$ of $0.500\ \text{mol dm}^{-3}$ sodium hydroxide. The acid also reacts with magnesium to produce hydrogen.

Which statements can be deduced?

1. The relative molecular mass of the acid is 98.
2. One mole of the acid can supply three moles of H^+ in complete neutralisation.
3. The acid is a strong acid.

- **A** none
- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

The molar mass is

$$M_r = \frac{29.4}{0.300} = 98,$$

so statement 1 is true.

Moles of acid in $25.0\ cm^3$:

$$0.300(0.0250) = 0.00750.$$

Moles of $NaOH$:

$$0.500(0.0450) = 0.0225.$$

The ratio is

$$\frac{0.0225}{0.00750} = 3,$$

so statement 2 is true.

Reaction with magnesium does not show whether the acid is fully ionised in water, so statement 3 cannot be deduced.

The answer is **E: 1 and 2 only**.

C-10

Back titration and purity

Answer F

QUESTION 10

A $2.10\ g$ impure sample of magnesium carbonate contains impurities that do not react with acids.

The sample is reacted with $100\ cm^3$ of $0.500\ mol\ dm^{-3}$ hydrochloric acid.

The excess hydrochloric acid then requires $20.0\ cm^3$ of $0.500\ mol\ dm^{-3}$ sodium hydroxide for complete neutralisation.

$$MgCO_3 + 2HCl \rightarrow MgCl_2 + CO_2 + H_2O$$

$$M_r(MgCO_3) = 84$$

What is the percentage by mass of magnesium carbonate in the original sample?

A 40%

B 50%

C 60%

D 70%

E 90%

F 80%

WORKED SOLUTION

Initial HCl:

$$0.500(0.100) = 0.0500 \text{ mol.}$$

HCl remaining is equal to the amount of NaOH used:

$$0.500(0.0200) = 0.0100 \text{ mol.}$$

Therefore HCl consumed by $MgCO_3$ is

$$0.0500 - 0.0100 = 0.0400 \text{ mol.}$$

From the 1:2 ratio,

$$n(MgCO_3) = 0.0200 \text{ mol.}$$

Mass of pure $MgCO_3$:

$$0.0200(84) = 1.68 \text{ g.}$$

Hence

$$\% \text{ purity} = \frac{1.68}{2.10} \times 100 = \boxed{80\%}.$$

The answer is **F**.

C-11

Molecular formula from combustion

Answer C

QUESTION 11

A compound contains carbon, hydrogen and oxygen only.

Complete combustion of 0.880 g of the compound produces 1.760 g of CO_2 and 0.720 g of H_2O.

The relative molecular mass of the compound is 88.

$$A_r: \quad C = 12,\ H = 1,\ O = 16$$

What is the molecular formula?

A CH_2O

B C_2H_4O

C $C_4H_8O_2$

D $C_2H_4O_2$

E $C_3H_6O_2$

F C_4H_8O

WORKED SOLUTION

Carbon:

$$n(CO_2) = \frac{1.760}{44} = 0.0400,$$

so there are 0.0400 mol C atoms.

Hydrogen:

$$n(H_2O) = \frac{0.720}{18} = 0.0400,$$

so there are 0.0800 mol H atoms.

Mass of carbon is $0.0400(12) = 0.480$ g; mass of hydrogen is 0.0800 g. Therefore oxygen mass is

$$0.880 - 0.480 - 0.080 = 0.320 \text{ g},$$

giving

$$n(O) = \frac{0.320}{16} = 0.0200.$$

The ratio is

$$C : H : O = 0.040 : 0.080 : 0.020 = 2 : 4 : 1.$$

Empirical formula: C_2H_4O, mass 44. Since $88 = 2(44)$, the molecular formula is

$$\boxed{C_4H_8O_2}.$$

The answer is **C**.

C-12

Composition of air

Answer **B**

QUESTION 12

A sample of dry air contains exactly 30.0 mol of gas.

Gas X contributes 3.78×10^{24} molecules to the sample.

Gas Y, if isolated at room temperature and pressure, would occupy 561.6 dm^3.

Take $N_A = 6.00 \times 10^{23}$ and molar gas volume at rtp as $24.0 \text{ dm}^3 \text{ mol}^{-1}$.

Which row is consistent with the approximate composition of dry air?

option	X	Y	remaining gases / mol
A	N_2	O_2	0.30
B	O_2	N_2	0.30
C	Ar	N_2	6.30
D	O_2	Ar	23.4
E	N_2	Ar	6.30

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

For X:

$$n = \frac{3.78 \times 10^{24}}{6.00 \times 10^{23}} = 6.30 \text{ mol.}$$

That is

$$\frac{6.30}{30.0} = 0.21 = 21\%,$$

so X is oxygen.

For Y:

$$n = \frac{561.6}{24.0} = 23.4 \text{ mol,}$$

which is

$$\frac{23.4}{30.0} = 0.78 = 78\%,$$

so Y is nitrogen.

Remaining gases:

$$30.0 - 6.30 - 23.4 = 0.30 \text{ mol}.$$

Therefore the correct row is **B**.

C-13 Two-dimensional chromatography **Answer E**

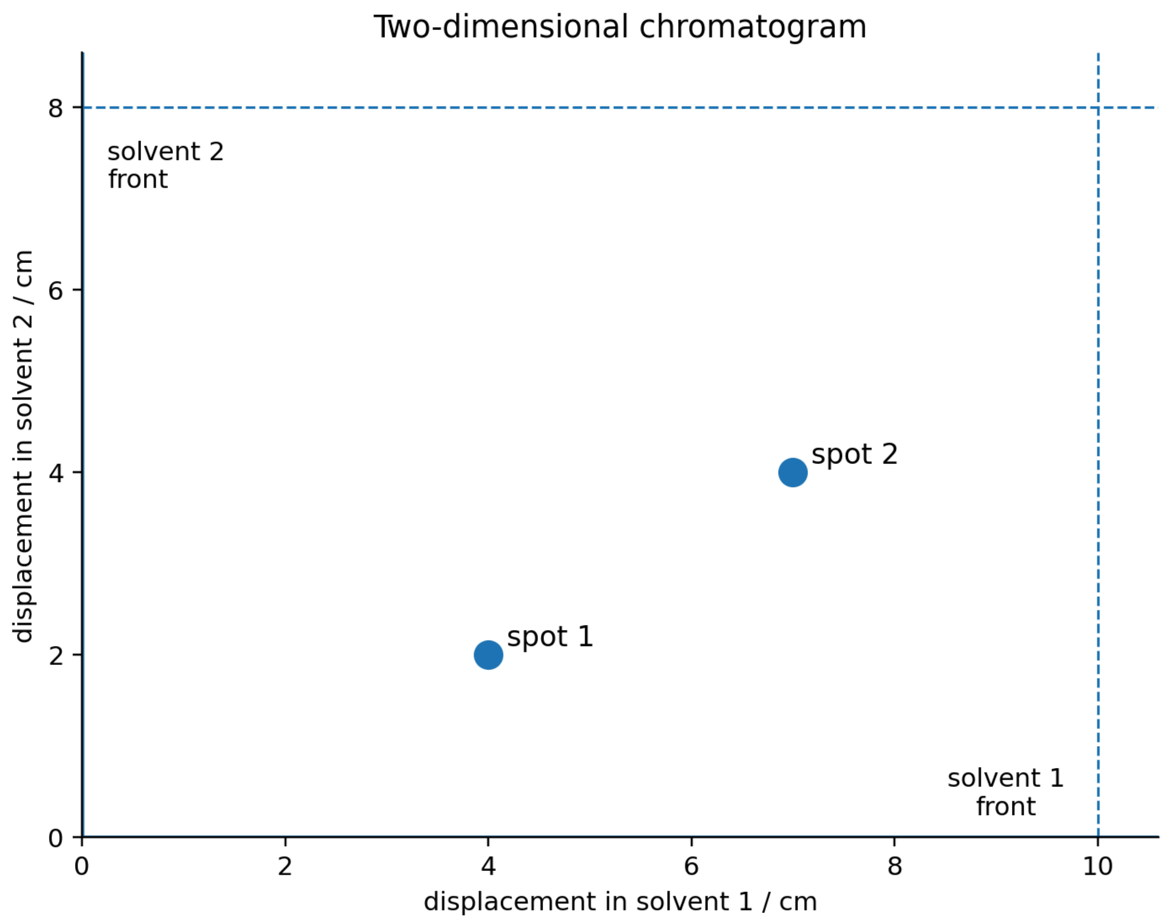
QUESTION 13

Four compounds have the following $\mathrm{R_f}$ values in two solvents.

compound	$\mathrm{R_f}$ in solvent 1	$\mathrm{R_f}$ in solvent 2
A	0.20	0.75
B	0.40	0.25
C	0.40	0.75
D	0.70	0.50

A mixture is analysed by two-dimensional chromatography. In the first direction, solvent 1 travels 10.0 cm. The paper is dried, rotated through 90°, and solvent 2 travels 8.0 cm.

The final chromatogram is shown.



Which compounds were present?

A A and B

B A and C

C B and C

D C and D

E B and D

WORKED SOLUTION

For spot 1:

$$R_{f,1} = \frac{4.0}{10.0} = 0.40, \qquad R_{f,2} = \frac{2.0}{8.0} = 0.25.$$

This identifies compound B.

For spot 2:

$$R_{f,1} = \frac{7.0}{10.0} = 0.70, \qquad R_{f,2} = \frac{4.0}{8.0} = 0.50.$$

This identifies compound D.

Therefore the mixture contained **B and D**, answer **E**.

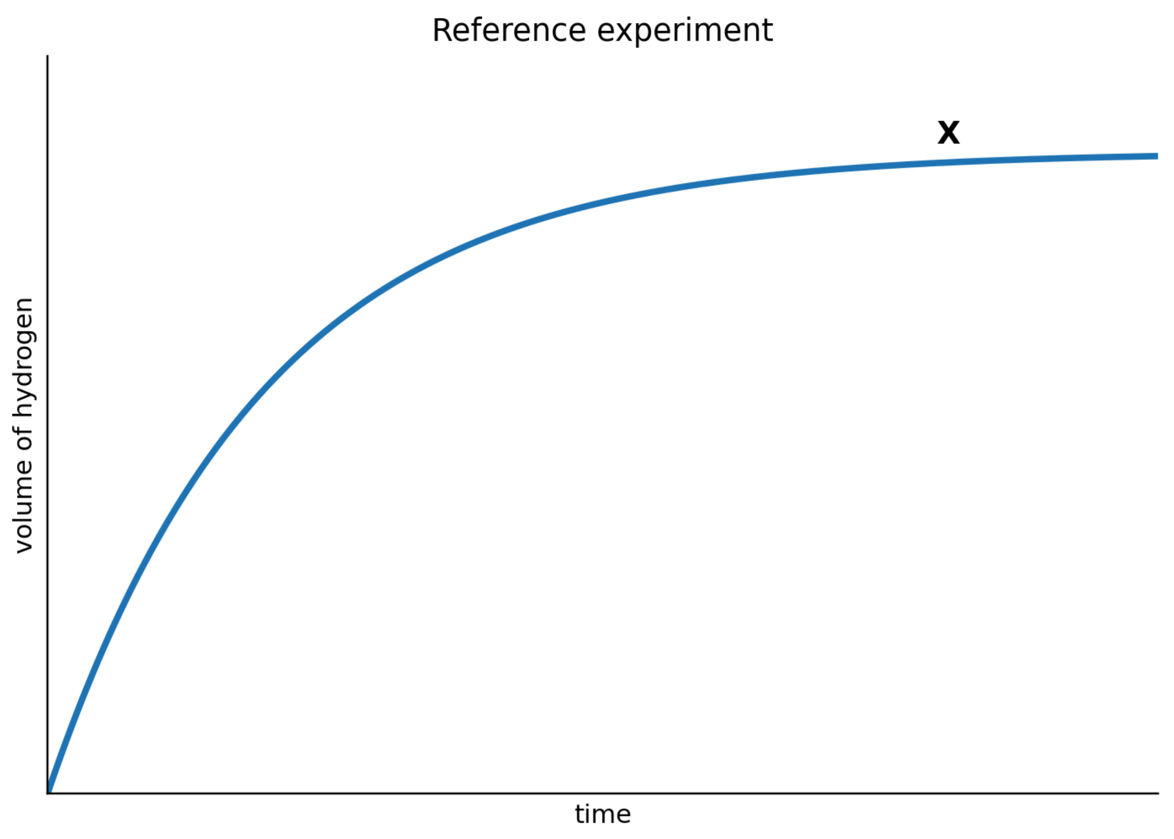
C-14

Rate and limiting reagent

Answer A

QUESTION 14

Curve X is obtained when $0.240\ \mathrm{g}$ of magnesium granules react with $100\ \mathrm{cm}^3$ of $0.100\ \mathrm{mol\ dm}^{-3}$ hydrochloric acid.



The experiment is repeated using:

- the same mass of magnesium, but powdered;
- $100\ \mathrm{cm}^3$ of $0.150\ \mathrm{mol\ dm}^{-3}$ hydrochloric acid.

$$A_r(Mg) = 24$$

Which statement correctly compares the second curve with X?

option	initial gradient	final volume of H_2
A	greater	$1.5 \times X$
B	greater	same as X
C	same	$1.5 \times X$
D	smaller	$1.5 \times X$
E	greater	$2.0 \times X$

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

Moles of magnesium:

$$\frac{0.240}{24} = 0.0100.$$

Complete reaction would need 0.0200 mol HCl.

First experiment:

$$n(HCl) = 0.100(0.100) = 0.0100 \text{ mol},$$

so acid is limiting.

Second experiment:

$$n(HCl) = 0.150(0.100) = 0.0150 \text{ mol},$$

which is still limiting. Therefore the final hydrogen amount is larger by

$$\frac{0.0150}{0.0100} = 1.5.$$

The second experiment also has a higher acid concentration and a larger magnesium surface area, so its initial rate is greater.

The correct row is **A**.

C-15

Equilibrium from observations

Answer C

QUESTION 15

Gaseous reactants P and Q establish equilibrium with gaseous products R and S.

During establishment of equilibrium from the reactants:

- for every 0.10 mol of P consumed,
- 0.20 mol of Q is consumed,
- 0.10 mol of R is formed.

Increasing pressure changes the rate at which equilibrium is reached but does **not** change the equilibrium yield.

Increasing temperature reduces the equilibrium yield of products.

Which equation is consistent with all the observations?

A $P + 2Q \rightleftharpoons R + S \qquad \Delta H < 0$

B $P + 2Q \rightleftharpoons 2R + S \qquad \Delta H > 0$

C $P + 2Q \rightleftharpoons R + 2S \qquad \Delta H < 0$

D $2P + Q \rightleftharpoons R + 2S \qquad \Delta H < 0$

E $P + 2Q \rightleftharpoons R + 2S \qquad \Delta H > 0$

F $P + Q \rightleftharpoons R + S \qquad \Delta H < 0$

WORKED SOLUTION

The observed mole changes give the ratio

$$P : Q : R = 1 : 2 : 1.$$

So the equation must begin

$$P + 2Q \rightleftharpoons R + \cdots$$

There are three gaseous reactant particles. Because pressure does not shift the equilibrium yield, the product side must also contain three gaseous particles. Thus the stoichiometry is

$$P + 2Q \rightleftharpoons R + 2S.$$

Increasing temperature reduces product yield, so the forward reaction is exothermic: $\Delta H < 0$.

The answer is **C**.

C-16

Bond energies

Answer **D**

QUESTION 16

Ethene reacts with hydrogen:

$$C_2H_4 + H_2 \rightarrow C_2H_6$$

The enthalpy change is $-125\ \mathrm{kJ\ mol^{-1}}$.

Mean bond energies are:

bond	bond energy / kJ mol^{-1}
H–H	436
C=C	612
C–C	348

What is the mean C–H bond energy implied by these data?

A 356 kJ mol^{-1}

B 388 kJ mol^{-1}

C 436 kJ mol^{-1}

D 413 kJ mol^{-1}

E 475 kJ mol^{-1}

WORKED SOLUTION

Relative to the reactants, hydrogenation breaks one C=C bond and one H–H bond, and forms one C–C bond and two additional C–H bonds.

Let the mean C–H bond energy be x. Then

$$-125 = (612 + 436) - (348 + 2x).$$

So

$$-125 = 700 - 2x \Rightarrow 2x = 825 \Rightarrow x = 412.5.$$

Therefore the closest value is $\boxed{413 \text{ kJ mol}^{-1}}$, answer **D**.

C-17

Calorimetry and limiting reagent

Answer C

QUESTION 17

$40.0\ \mathrm{cm}^3$ of $1.50\ \mathrm{mol\ dm}^{-3}$ hydrochloric acid is mixed with $60.0\ \mathrm{cm}^3$ of $0.500\ \mathrm{mol\ dm}^{-3}$ sodium hydroxide.

The temperature rises by $3.4°\mathrm{C}$.

Assume:

- the solution volumes are additive;
- density of the final solution = $1.0\ \mathrm{g\ cm}^{-3}$;
- specific heat capacity = $4.2\ \mathrm{J\ g}^{-1}\ °\mathrm{C}^{-1}$;
- only 85% of the energy released by the reaction is retained by the solution.

What is the molar enthalpy change for the neutralisation reaction?

A $-28\ \mathrm{kJ\ mol}^{-1}$

B $-40\ \mathrm{kJ\ mol}^{-1}$

C $-56\ \mathrm{kJ\ mol}^{-1}$

D $-48\ \mathrm{kJ\ mol}^{-1}$

E $-66\ \mathrm{kJ\ mol}^{-1}$

WORKED SOLUTION

Moles HCl:

$$1.50(0.0400) = 0.0600.$$

Moles NaOH:

$$0.500(0.0600) = 0.0300.$$

NaOH is limiting, so 0.0300 mol is neutralised.

The solution mass is 100 g. Energy retained:

$$q = mc\Delta T = 100(4.2)(3.4) = 1428\ \text{J}.$$

This is 85% of the energy released, so

$$q_{\text{released}} = \frac{1428}{0.85} = 1680\ \text{J} = 1.680\ \text{kJ}.$$

Per mole:

$$\frac{1.680}{0.0300} = 56.0\ \text{kJ mol}^{-1}.$$

Neutralisation is exothermic, so $\boxed{-56.0\ \text{kJ mol}^{-1}}$, answer **C**.

C-18

Metal extraction

Answer **B**

QUESTION 18

An iron ore contains 80.0% by mass of hydrated iron(III) oxide,

$$Fe_2O_3 \cdot 3H_2O.$$

The formula $Fe_2O_3 \cdot 3H_2O$ means that each formula unit contains one Fe_2O_3 unit together with three H_2O units.

Assume all the iron in the hydrated oxide can be extracted.

$$A_r: \quad Fe = 56,\ O = 16,\ H = 1$$

What is the minimum mass of ore required to produce 56.0 tonnes of iron?

A 100 tonnes

B 134 tonnes

C 112 tonnes

D 125 tonnes

E 160 tonnes

F 214 tonnes

WORKED SOLUTION

The relative formula mass of $Fe_2O_3 \cdot 3H_2O$ is

$$2(56) + 3(16) + 3(18) = 214.$$

Each 214 mass units contain 112 mass units of iron, so the iron fraction in pure hydrated oxide is $112/214$.

The ore is only 80.0% hydrated oxide, so the iron fraction of the ore is

$$0.80\left(\frac{112}{214}\right).$$

If m tonnes of ore are needed,

$$m \cdot 0.80\left(\frac{112}{214}\right) = 56.0.$$

Thus

$$m = 133.75 \text{ tonnes} \approx \boxed{134 \text{ tonnes}}.$$

The answer is **B**.

C-19

Electrolysis and concentration

Answer **E**

QUESTION 19

$250\ \mathrm{cm^3}$ of $1.00\ \mathrm{mol\ dm^{-3}}$ copper(II) sulfate solution is electrolysed using inert electrodes.

After some time, $6.40\ \mathrm{g}$ of copper has been deposited at the cathode.

Water is then added so that the solution volume is again exactly $250\ \mathrm{cm}^3$.

$$A_r(\mathrm{Cu}) = 64$$

What is the final concentration of Cu^{2+}?

A $0.20\ \mathrm{mol\ dm}^{-3}$

B $0.40\ \mathrm{mol\ dm}^{-3}$

C $0.50\ \mathrm{mol\ dm}^{-3}$

D $0.75\ \mathrm{mol\ dm}^{-3}$

E $0.60\ \mathrm{mol\ dm}^{-3}$

F $0.90\ \mathrm{mol\ dm}^{-3}$

WORKED SOLUTION

Initial amount of Cu^{2+}:

$$1.00(0.250) = 0.250\ \mathrm{mol}.$$

Copper deposited:

$$n(\mathrm{Cu}) = \frac{6.40}{64} = 0.100\ \mathrm{mol}.$$

Therefore 0.100 mol of Cu^{2+} has been removed, leaving

$$0.250 - 0.100 = 0.150\ \mathrm{mol}.$$

At a final volume of $0.250\ \mathrm{dm}^3$,

$$[\mathrm{Cu}^{2+}] = \frac{0.150}{0.250} = \boxed{0.600\ \mathrm{mol\ dm}^{-3}}.$$

The answer is **E**.

C-20

Organic structure from constraints

Answer **B**

QUESTION 20

A hydrocarbon X:

- produces $120\ dm^3$ of carbon dioxide when one mole is completely burned;
- reacts with exactly one mole of hydrogen per mole of X;
- undergoes addition polymerisation;
- has a terminal C=C bond;
- forms a **branched saturated hydrocarbon** on hydrogenation.

At rtp, one mole of gas occupies $24\ dm^3$.

The candidate structures are shown as condensed structural formulae.

structure	condensed structural formula
A	$CH_2 = CHCH_2CH_2CH_3$
B	$CH_2 = C(CH_3)CH_2CH_3$
C	$CH_3CH = CHCH_2CH_3$
D	$CH_3C(CH_3) = CHCH_3$
E	cyclopentane, $(CH_2)_5$ ring

Which could be X?

- **A** structure A
- **B** structure B
- **C** structure C
- **D** structure D

E structure E

WORKED SOLUTION

The carbon dioxide volume corresponds to

$$\frac{120}{24} = 5\ \mathrm{mol\ CO_2},$$

so X contains five carbon atoms.

Reaction with exactly one mole of hydrogen indicates one C=C bond. Addition polymerisation also requires an alkene.

The double bond must be terminal, leaving A or B. Hydrogenating A gives unbranched pentane, whereas hydrogenating B gives branched 2-methylbutane.

Therefore X is **structure B**.

C-21

Multiple double bonds

Answer A

QUESTION 21

A hydrocarbon contains two carbon-carbon double bonds.

A $0.100\ \mathrm{cm}^3$ sample has density $0.820\ \mathrm{g\ cm}^{-3}$ and relative molecular mass 82.

What is the minimum volume of $0.250\ \mathrm{mol\ dm}^{-3}$ bromine solution required to react completely with the hydrocarbon?

A $8.0\ \mathrm{cm}^3$

B $2.0\ \mathrm{cm}^3$

C $4.0\ \mathrm{cm}^3$

D $6.0\ \mathrm{cm}^3$

E $16.0\ \text{cm}^3$

WORKED SOLUTION

Mass of hydrocarbon:

$$0.100(0.820) = 0.0820\ \text{g}.$$

Moles:

$$\frac{0.0820}{82} = 0.00100\ \text{mol}.$$

There are two C=C bonds, so each mole consumes two moles of bromine. Thus

$$n(Br_2) = 0.00200\ \text{mol}.$$

Required volume:

$$V = \frac{0.00200}{0.250} = 0.00800\ \text{dm}^3 = \boxed{8.00\ \text{cm}^3}.$$

The answer is **A**.

C-22

Carboxylic acid and esterification

Answer **D**

QUESTION 22

A saturated monocarboxylic acid X reacts with propan-1-ol to form an ester Y.

$$M_r(Y) = 130$$

and

$$M_r(\text{propan-1-ol}) = 60.$$

What is the molecular formula of X?

A $C_2H_4O_2$

B $C_3H_6O_2$

C $C_5H_{10}O_2$

D $C_4H_8O_2$

E $C_6H_{12}O_2$

WORKED SOLUTION

Esterification gives

$$\text{acid} + \text{alcohol} \rightarrow \text{ester} + H_2O.$$

Therefore

$$M_r(X) + 60 = 130 + 18,$$

so

$$M_r(X) = 88.$$

A saturated monocarboxylic acid has formula $C_nH_{2n}O_2$, so

$$12n + 2n + 32 = 88.$$

Thus

$$14n = 56 \Rightarrow n = 4.$$

The formula is $\boxed{C_4H_8O_2}$, answer **D**.

C-23

Combustion and cracking

Answer F

QUESTION 23

$20\ \text{cm}^3$ of a gaseous alkane X requires $190\ \text{cm}^3$ of oxygen for complete combustion.

Both gas volumes are measured at the same temperature and pressure.

X can crack according to

$$X \rightarrow C_2H_4 + Y$$

with one molecule of Y formed for each molecule of X.

What is Y?

A C_2H_6

B C_3H_6

C C_3H_8

D C_4H_8

E C_5H_{10}

F C_4H_{10}

WORKED SOLUTION

For an alkane C_nH_{2n+2}, complete combustion requires

$$n + \frac{2n + 2}{4} = \frac{3n + 1}{2}$$

moles of oxygen per mole of alkane.

The gas-volume ratio is

$$\frac{190}{20} = 9.5.$$

Hence

$$\frac{3n + 1}{2} = 9.5 \Rightarrow 3n + 1 = 19 \Rightarrow n = 6.$$

So $X = C_6H_{14}$. Balancing

$$C_6H_{14} \rightarrow C_2H_4 + Y$$

gives

$$Y = \boxed{C_4H_{10}}.$$

The answer is **F**.

C-24

Water treatment and gas quantities

Answer **D**

QUESTION 24

A water-treatment plant processes $480\,000\ \mathrm{dm}^3$ of water each day.

Chlorine is added at a rate of $0.355\ \mathrm{mg\ dm}^{-3}$.

Assume all the chlorine is supplied as $Cl_2(g)$.

$$A_r(Cl) = 35.5$$

and one mole of gas occupies $24.0\ \mathrm{dm}^3$ at rtp.

What volume of chlorine gas is used each day?

A $24.0\ \mathrm{dm}^3$

B $36.0\ \mathrm{dm}^3$

C $48.0\ \mathrm{dm}^3$

D $57.6\ \mathrm{dm}^3$

E $71.0\ \mathrm{dm}^3$

F $96.0\ \mathrm{dm}^3$

WORKED SOLUTION

Mass of chlorine used:

$$480000(0.355) = 170400 \text{ mg} = 170.4 \text{ g}.$$

Since $M_r(Cl_2) = 71$,

$$n(Cl_2) = \frac{170.4}{71} = 2.40 \text{ mol}.$$

At rtp,

$$V = 2.40(24.0) = \boxed{57.6 \text{ dm}^3}.$$

The answer is **D**.

C-25

Qualitative analysis

Answer F

QUESTION 25

An aqueous salt X gives:

- a **green precipitate** on addition of aqueous sodium hydroxide;
- a **white precipitate** on addition of acidified aqueous barium chloride.

Which is the most likely identity of X?

A $FeCl_2$

B $Fe_2(SO_4)_3$

C $CuSO_4$

D $MgSO_4$

E Na_2SO_4

F $FeSO_4$

WORKED SOLUTION

A green hydroxide precipitate identifies Fe^{2+}.

A white precipitate with acidified barium chloride identifies sulfate ions, SO_4^{2-}.

The salt must therefore be $\boxed{FeSO_4}$, answer **F**.

C-26

Multi-stage gas stoichiometry

Answer E

QUESTION 26

Potassium chlorate decomposes completely:

$$2KClO_3 \rightarrow 2KCl + 3O_2$$

All the oxygen produced is then used to oxidise excess carbon monoxide:

$$2CO + O_2 \rightarrow 2CO_2.$$

A $24.5\ g$ sample of pure potassium chlorate is used.

$$M_r(KClO_3) = 122.5$$

and one mole of gas occupies $24.0\ dm^3$ at rtp.

What volume of carbon dioxide is ultimately formed?

- **A** $4.8\ dm^3$
- **B** $7.2\ dm^3$
- **C** $9.6\ dm^3$
- **D** $12.0\ dm^3$
- **E** $14.4\ dm^3$

F $18.0\ dm^3$

WORKED SOLUTION

Moles of potassium chlorate:

$$\frac{24.5}{122.5} = 0.200 \text{ mol.}$$

From

$$2KClO_3 \rightarrow 3O_2,$$

the oxygen amount is

$$0.200\left(\frac{3}{2}\right) = 0.300 \text{ mol.}$$

Then

$$O_2 : CO_2 = 1 : 2,$$

so

$$n(CO_2) = 0.600 \text{ mol.}$$

Volume:

$$0.600(24.0) = \boxed{14.4\ dm^3}.$$

The answer is **E**.

C-27

Mixture analysis and back titration

Answer A

QUESTION 27

A 3.00 g mixture contains only magnesium and magnesium oxide.

It is reacted with $100\ cm^3$ of $2.00\ mol\ dm^{-3}$ hydrochloric acid.

The excess hydrochloric acid requires $20.0\ cm^3$ of $1.00\ mol\ dm^{-3}$ sodium hydroxide for complete neutralisation.

$$Mg + 2HCl \rightarrow MgCl_2 + H_2$$

$$MgO + 2HCl \rightarrow MgCl_2 + H_2O$$

$$A_r(Mg) = 24, \qquad M_r(MgO) = 40.$$

What volume of hydrogen gas is produced at rtp?

$$1 \text{ mol gas} = 24.0 \text{ dm}^3.$$

A 0.90 dm^3

B 0.45 dm^3

C 0.60 dm^3

D 0.75 dm^3

E 1.20 dm^3

F 1.80 dm^3

WORKED SOLUTION

Initial HCl:

$$0.100(2.00) = 0.200 \text{ mol}.$$

Excess HCl:

$$0.0200(1.00) = 0.0200 \text{ mol}.$$

So HCl consumed by the mixture is

$$0.200 - 0.0200 = 0.180 \text{ mol}.$$

Both Mg and MgO consume two moles HCl per mole, so

$$n(Mg) + n(MgO) = \frac{0.180}{2} = 0.0900.$$

Let $n(Mg) = x$ and $n(MgO) = y$. Then

$$x + y = 0.0900$$

and

$$24x + 40y = 3.00.$$

Using $x = 0.0900 - y$:

$$24(0.0900 - y) + 40y = 3.00$$

$$2.16 + 16y = 3.00 \Rightarrow y = 0.0525.$$

Hence

$$x = 0.0900 - 0.0525 = 0.0375.$$

Only magnesium metal produces hydrogen, in a 1:1 ratio, so

$$n(H_2) = 0.0375.$$

Therefore

$$V(H_2) = 0.0375(24.0) = \boxed{0.900\ dm^3}.$$

The answer is **A**.

LEVEL 2 · HARDER THAN ESAT

Biology

27 QUESTIONS · COMPLETE SOLUTIONS

Preserved Biology Harder than ESAT source set. Question wording, option order, answer and solution method are unchanged.

B-01

Cells · surface area to volume

Answer C

QUESTION 1

Two unicellular organisms, X and Y, are approximately cube-shaped.

The side length of Y is twice the side length of X.

Both organisms have:

- the same oxygen demand per unit volume;
- the same maximum rate of oxygen diffusion per unit surface area.

Compared with X, by what factor is the **oxygen demand per unit surface area** of Y greater?

A 1/2

B 1

C 2

D 4

E 8

WORKED SOLUTION

For a cube, volume is $\mathbf{s^3}$ and surface area is $\mathbf{6s^2}$.

$$\text{oxygen demand per unit surface area} \propto s^3 / 6s^2 = s/6$$

Doubling the side length therefore doubles the oxygen demand per unit surface area.

Answer: 2.

B-02

Movement across membranes

Answer F

QUESTION 2

A section of dialysis tubing contains a solution of glucose and starch. It is placed in a large beaker of pure water.

The membrane is permeable to water and glucose, but not to starch.

Consider the statements while the system is approaching equilibrium.

1. While the glucose concentration remains higher inside the tubing than outside, there is a net movement of glucose out of the tubing.
2. Starch can leave the tubing by osmosis because its concentration is higher inside.
3. Even after the glucose concentrations have become equal, water may still show a net movement into the tubing because starch remains inside.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Glucose can cross the membrane, so while its concentration is higher inside, its net movement is outward by diffusion.

Osmosis describes the net movement of **water**, not starch, so statement 2 is false.

Because starch cannot cross the membrane, it can continue to lower the water potential inside the tubing. Water may therefore continue to enter even after glucose has equilibrated.

Statements 1 and 3 only are correct.

B-03 Diffusion and active transport Answer **B**

QUESTION 3

A root-hair cell contains nitrate ions at a concentration of **4 mmol dm^{-3}**.

A respiratory poison is added which completely stops active transport, but the cell membrane remains permeable to nitrate ions.

The cell is placed separately into three soil solutions:

soil solution	nitrate concentration / mmol dm^{-3}
X	1
Y	4
Z	10

Which row gives the expected **net passive movement of nitrate ions immediately after active transport stops**?

A X: into cell; Y: no net movement; Z: out of cell

B X: out of cell; Y: no net movement; Z: into cell

C X: out of cell; Y: into cell; Z: into cell

D X: no net movement; Y: no net movement; Z: into cell

E X: out of cell; Y: out of cell; Z: into cell

WORKED SOLUTION

With active transport disabled, only passive movement down the concentration gradient remains.

- X: 1 outside versus 4 inside → net movement **out**.
- Y: 4 outside versus 4 inside → **no net movement**.
- Z: 10 outside versus 4 inside → net movement **in**.

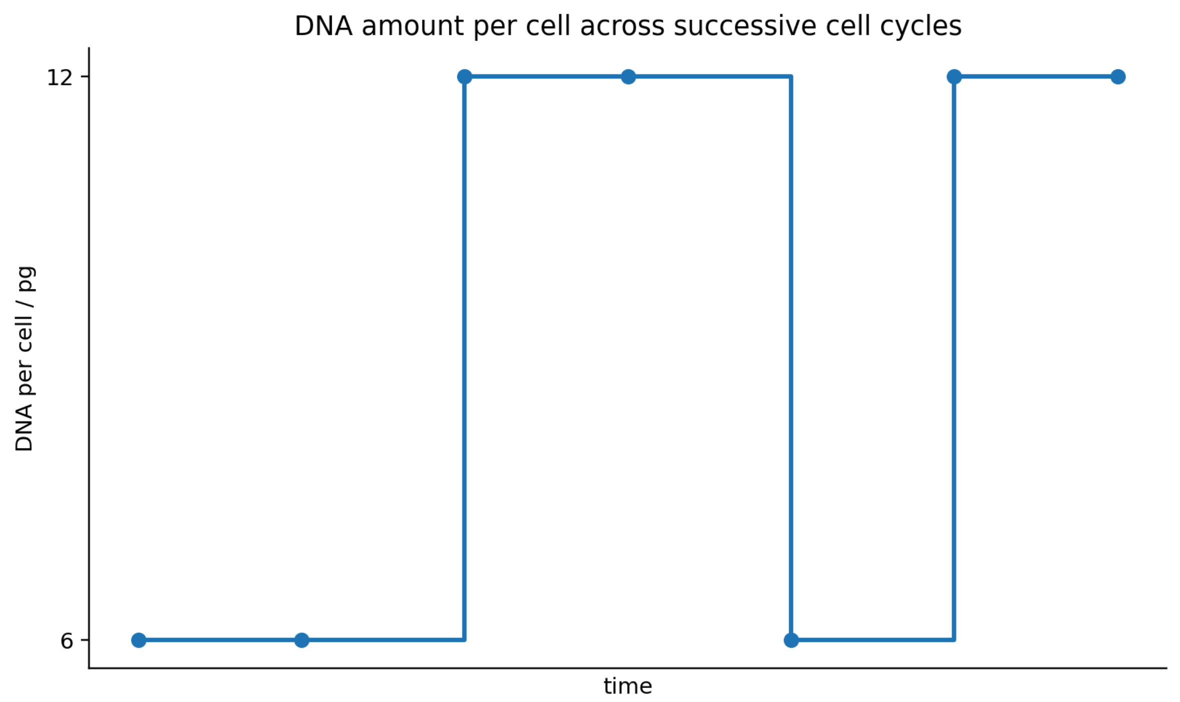
Answer B.

B-04

Cell cycle · DNA and chromosomes

Answer **C**

QUESTION 4



A diploid species has **12 chromosomes** in each body cell.

Immediately after mitosis, one cell contains **6 pg** of DNA.

It passes through another cell cycle, as represented by the first rise and fall on the graph. Both daughter cells then replicate their DNA and are observed at the **second high plateau**, immediately before their next mitosis.

Which row is correct at this second high plateau?

A 1 cell; 12 total chromosomes; 12 pg total DNA

B 2 cells; 24 total chromosomes; 12 pg total DNA

C 2 cells; 24 total chromosomes; 24 pg total DNA

D 2 cells; 48 total chromosomes; 24 pg total DNA

E 4 cells; 48 total chromosomes; 24 pg total DNA

WORKED SOLUTION

After the first mitosis there are two cells. Each has 12 chromosomes and 6 pg DNA.

DNA replication doubles DNA quantity but does not change chromosome number.

So immediately before the next mitosis, each of the two cells has 12 chromosomes and 12 pg DNA.

$$\text{across both cells: 24 chromosomes and 24 pg DNA}$$

Answer C.

B-05

Meiosis and sex determination

Answer C

QUESTION 5

A male mammal has a diploid chromosome number of **2n = 10** and an XY sex-chromosome system.

One diploid germ cell completes meiosis normally and produces four sperm.

Across **all four sperm together**, which row is correct?

A 8 total autosomes; 4 total sex chromosomes; X:Y sperm = 2:2

B 16 total autosomes; 2 total sex chromosomes; X:Y sperm = 1:1

C 16 total autosomes; 4 total sex chromosomes; X:Y sperm = 2:2

D 20 total autosomes; 4 total sex chromosomes; X:Y sperm = 1:3

E 16 total autosomes; 8 total sex chromosomes; X:Y sperm = 2:2

WORKED SOLUTION

Each sperm is haploid and therefore contains 5 chromosomes: 4 autosomes plus 1 sex chromosome.

$$4 \text{ sperm} \times 4 \text{ autosomes} = 16 \text{ autosomes}; 4 \text{ sperm} \times 1 \text{ sex chromosome} = 4 \text{ sex chromosomes}$$

Normal meiosis of an XY germ cell gives equal numbers of X-bearing and Y-bearing sperm, so the ratio is 2:2.

Answer C.

B-06

Inheritance · conditional probability

Answer **D**

QUESTION 6

In an animal species:

- allele A is dominant and produces black fur in genotype Aa;
- genotype aa produces white fur;
- genotype AA is lethal before birth.

Two **living black animals** mate.

What is the probability that **exactly one of their next two live offspring is white**?

A 1/9

B 2/9

C 1/3

D 4/9

E 2/3

WORKED SOLUTION

A living black animal cannot be AA because AA is lethal, so both parents are Aa.

Before lethality the cross gives 1/4 AA, 1/2 Aa and 1/4 aa. Among **live** offspring, the probabilities therefore become:

$$P(\text{black} \mid \text{alive}) = 2/3 \qquad P(\text{white} \mid \text{alive}) = 1/3$$

Exactly one white among two live offspring can occur in either order:

$$2 \times (1/3) \times (2/3) = 4/9$$

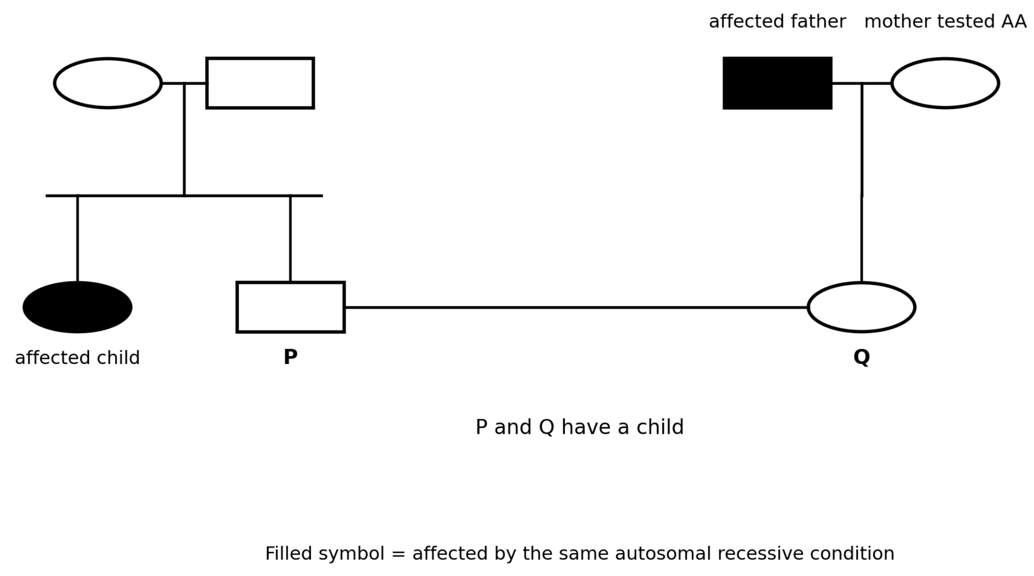
Answer D.

B-07

Pedigrees · conditional probability

Answer C

QUESTION 7



The same autosomal recessive condition occurs in both families shown.

In the family on the left, two unaffected parents have an affected child and an unaffected son, P.

In the family on the right, Q's father is affected and Q's unaffected mother has been genetically tested and is homozygous dominant.

P and Q have a child.

What is the probability that their child is affected?

A 1/12

B 1/8

C 1/6

D 1/4

E 1/3

WORKED SOLUTION

P's parents must both be Aa because they are unaffected but have an affected aa child.

Given that P is unaffected, P has a 2/3 probability of being Aa.

Q's father is aa and Q's mother is AA, so Q must be Aa.

If P is a carrier, Aa × Aa gives an affected child with probability 1/4.

$$(2/3) \times (1/4) = 1/6$$

Answer C.

B-08

DNA base pairing

Answer **D**

QUESTION 8

One strand of a DNA molecule has the following base composition:

base	percentage
A	18
T	30
C	22
G	30

The complete double-stranded DNA molecule contains **2000 bases**.

How many **guanine bases** are present in the complete molecule?

A 440

B 480

C 500

D 520

E 600

WORKED SOLUTION

On the complementary strand, the percentage of G equals the percentage of C on the first strand, so it is 22%.

Across the two equal-length strands, the overall fraction of G is therefore:

$$(30\% + 22\%) / 2 = 26\%$$

$$0.26 \times 2000 = 520$$

Answer D.

B-09

Mutation and the genetic code

Answer C

QUESTION 9

The following coding-DNA triplets have the stated meanings.

triplet	amino acid
GCT	alanine
GCC	alanine
GTT	valine
GAT	aspartic acid
CCT	proline

A gene originally contains **GCT-GAT-CCT**.

Two mutations occur separately:

X: GCC-GAT-CCT

Y: GTT-GAT-CCT

1. Mutation X must alter the amino-acid sequence.
2. Mutation Y alters one amino acid in the encoded sequence.
3. Mutation Y must necessarily produce a visible change in phenotype.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

GCT and GCC both code for alanine, so mutation X does not alter this amino acid.

GTT codes for valine rather than alanine, so mutation Y changes one amino acid.

An amino-acid change does not guarantee a visible phenotypic change.

Statement 2 only is correct — answer C.

B-10

Gene technologies | Answer B

QUESTION 10

A bacterial strain is initially sensitive to kanamycin.

Scientists use a plasmid containing a functional kanamycin-resistance gene. The plasmid is cut at a site **outside** this resistance gene.

A gene of interest is cut with the same restriction enzyme, mixed with the plasmid and DNA ligase, and the bacteria are transformed.

The bacteria are then grown on agar containing kanamycin. Some colonies grow.

Assume that no new mutation conferring kanamycin resistance occurs during the experiment.

Which conclusion is justified?

A Every growing colony must contain the gene of interest.

B Every growing colony must have obtained DNA carrying a functional kanamycin-resistance gene, but growth alone does not show whether the gene of interest was inserted.

C Every bacterium exposed to the plasmid must have been transformed.

D The gene of interest must have entered the bacterial chromosome rather than a plasmid.

E DNA ligase must have cut the plasmid DNA.

WORKED SOLUTION

The original bacteria are kanamycin-sensitive. A colony that grows on kanamycin must therefore have acquired functioning resistance under the conditions described.

However, a cut plasmid can potentially re-form without receiving the desired insert. Kanamycin survival therefore does not prove that the gene of interest is present.

Answer B.

B-11 Stem cells Answer **B**

QUESTION 11

Three human cell populations have these properties.

population	observed ability
P	can give rise to all cell types required to form a complete organism
Q	can differentiate into any body-cell type but cannot by itself form a complete organism
R	obtained from adult bone marrow and can form several types of blood cell

A patient requires replacement blood cells.

Which row is correct?

- **A** P pluripotent; Q totipotent; R multipotent; least broad sufficient population = P
- **B** P totipotent; Q pluripotent; R multipotent; least broad sufficient population = R
- **C** P totipotent; Q multipotent; R pluripotent; least broad sufficient population = Q
- **D** P pluripotent; Q multipotent; R totipotent; least broad sufficient population = R
- **E** P multipotent; Q pluripotent; R totipotent; least broad sufficient population = P

WORKED SOLUTION

P is totipotent because it can generate all cell types needed for a complete organism.

Q is pluripotent because it can generate all body-cell types but not a complete organism.

R is multipotent because it can form a restricted range of related cell types.

For replacement blood cells, R is the least developmentally broad population that could be sufficient.

Answer B.

B-12 Variation · genes and environment Answer **E**

QUESTION 12

Two genetically different clonal lines, A and B, are grown under two nitrate conditions. All other measured conditions are kept the same.

clonal line	low nitrate: mean height / cm	high nitrate: mean height / cm
A	12	20
B	16	24

1. Environmental conditions can affect height.
2. Genetic differences could contribute to differences in height.
3. These data prove that nitrate availability is the only environmental factor that can affect plant height.

Which conclusions are supported?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Within each clonal line, height changes when nitrate availability changes, supporting an environmental effect.

Clonal line B is taller than clonal line A under both nitrate conditions, which is consistent with a genetic contribution.

The experiment does not establish that nitrate is the only environmental factor capable of affecting height.

Statements 1 and 2 only are supported — answer E.

B-13

Enzymes · experimental interpretation

Answer F

QUESTION 13

An enzyme is tested at three temperatures and three pH values. The table shows the time required to produce the **same fixed quantity of product**.

	20°C	40°C	60°C
pH 4	180 s	60 s	>300 s
pH 6	120 s	30 s	240 s
pH 8	200 s	75 s	>300 s

1. Of the conditions tested, pH 6 and 40°C gives the highest reaction rate.
2. These results prove that the enzyme's exact optimum is pH 6 and exactly 40°C.
3. At pH 6, the relative reaction rate at 40°C is four times that at 20°C.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

For a fixed quantity of product, rate is inversely proportional to the time taken.

The shortest measured time is 30 s at pH 6 and 40°C, so statement 1 is true.

$$\text{relative rate ratio} = (1/30) / (1/120) = 4$$

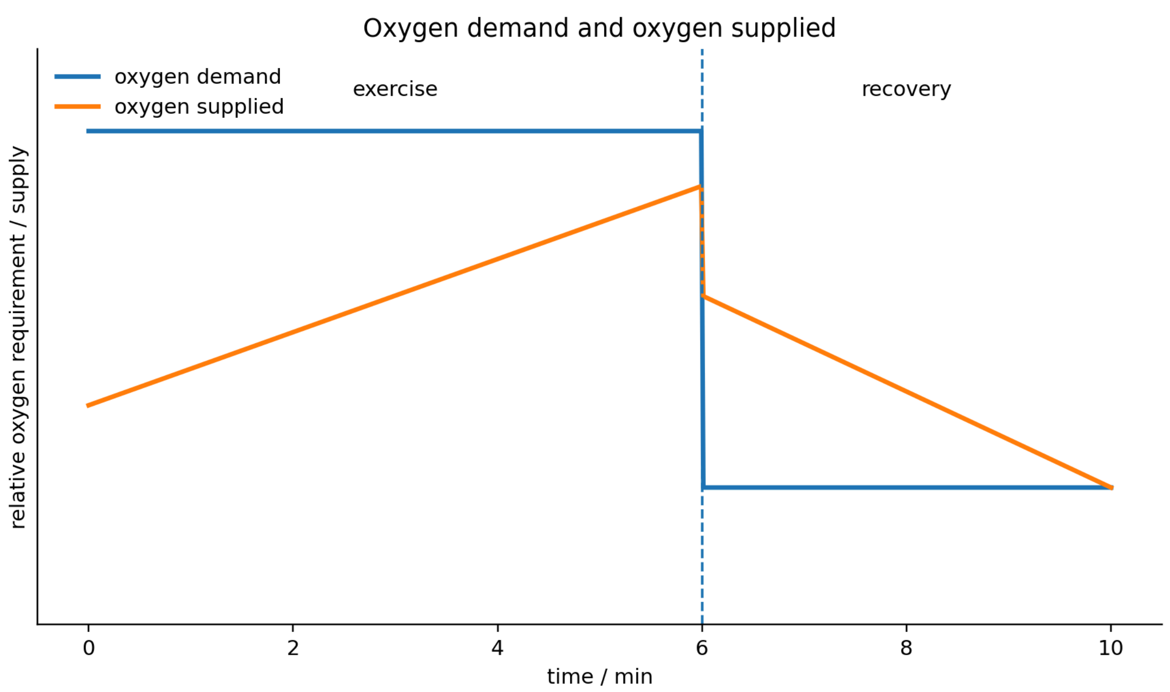
Statement 3 is therefore true.

The exact optimum could lie between the values tested, so statement 2 is not justified.

Answer F: 1 and 3 only.

B-14 Respiration during exercise Answer E

QUESTION 14



The graph shows relative oxygen demand and oxygen supplied during six minutes of exercise followed by recovery.

1. During the exercise period, the graph is consistent with anaerobic respiration contributing to energy release.
2. During the early recovery period, oxygen supply exceeding the current oxygen demand is consistent with repayment of oxygen debt.
3. When oxygen supply and demand become equal at the end of recovery, aerobic respiration must have stopped.

Which statements are correct?

- **A** none
- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

During exercise, oxygen demand exceeds the oxygen supplied, so aerobic respiration alone cannot account for the full stated demand. This is consistent with anaerobic respiration contributing.

During early recovery, oxygen supply remains above the current demand, which is consistent with repayment of oxygen debt.

Equal supply and demand at the end does not mean respiration has stopped; it means oxygen supply is meeting the current requirement.

Statements 1 and 2 only — answer E.

B-15

Gas exchange · quantitative diffusion

Answer C

QUESTION 15

A simple cubic organism has side length **s mm**.

Its cells consume oxygen at a rate equivalent to **1 unit per mm^3 per minute**.

Oxygen can cross the entire external surface at a maximum rate of **0.5 units per mm^2 per minute**.

The organism has no internal transport system.

What is the largest value of s for which diffusion across the surface could supply enough oxygen?

A 1 mm

B 2 mm

C 3 mm

D 4 mm

E 6 mm

WORKED SOLUTION

Oxygen demand equals the volume:

$$\text{demand} = s^3$$

Surface area is $6s^2$, so maximum supply is:

$$0.5 \times 6s^2 = 3s^2$$

Enough oxygen is supplied when $3s^2 \geq s^3$. For $s > 0$ this gives $s \leq 3$.

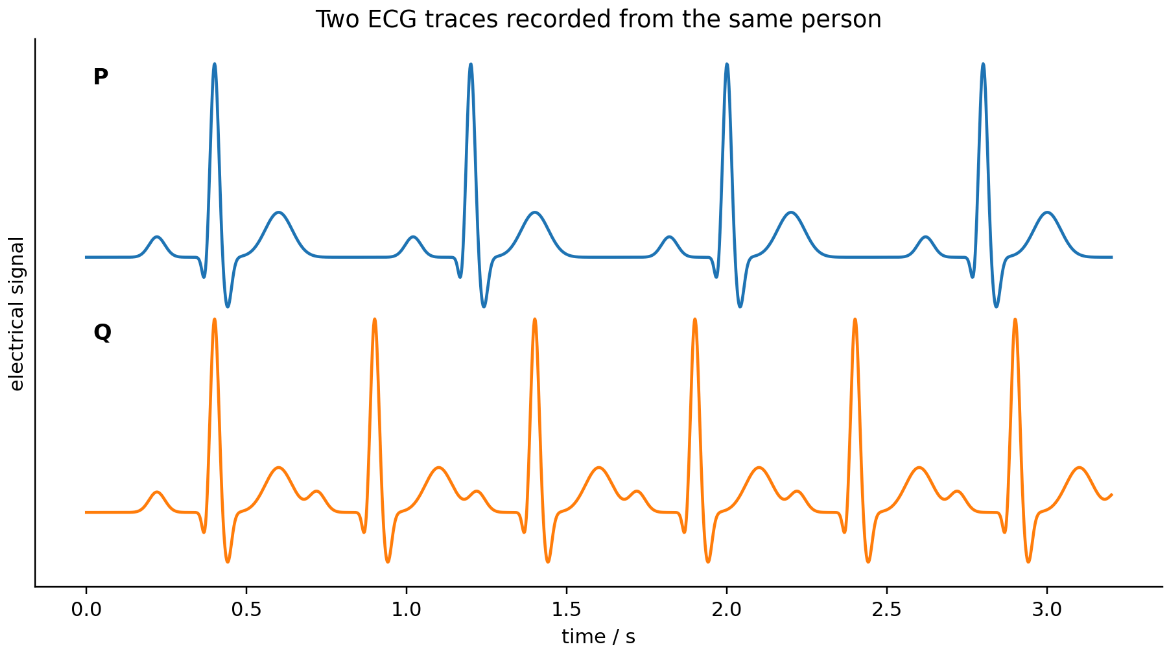
Largest possible side length = 3 mm — answer C.

B-16

ECG · heart rate and cardiac output

Answer D

QUESTION 16



Two ECG traces are recorded from the same person.

The spacing between corresponding large peaks is approximately **0.80 s in P** and **0.50 s in Q**.

Assume the volume of blood pumped per heartbeat is unchanged.

Which row is correct?

- **A** P = 60 min^{-1}; Q = 75 min^{-1}; blood/minute ratio Q:P = 1.25
- **B** P = 75 min^{-1}; Q = 100 min^{-1}; ratio = 1.33
- **C** P = 75 min^{-1}; Q = 120 min^{-1}; ratio = 1.50
- **D** P = 75 min^{-1}; Q = 120 min^{-1}; ratio = 1.60

E P = 80 min^{-1}; Q = 120 min^{-1}; ratio = 1.50

WORKED SOLUTION

$$\text{P heart rate} = 60 / 0.80 = 75 \text{ beats min}^{-1}$$

$$\text{Q heart rate} = 60 / 0.50 = 120 \text{ beats min}^{-1}$$

If stroke volume is unchanged, blood pumped per minute changes in the same ratio:

$$120 / 75 = 1.60$$

Answer D.

B-17 Kidney · blood composition **Answer B**

QUESTION 17

The concentrations of four substances are measured in two vessels connected to the same organ.

substance	vessel P	vessel Q
oxygen	9	6
carbon dioxide	5	7
urea	7	2
glucose	5	5

Values are in arbitrary but comparable units.

Which interpretation best fits the data?

A The organ is a lung; P is a pulmonary vein and Q a pulmonary artery.

B The organ is a kidney; P could be a renal artery and Q a renal vein.

C The organ is a liver; P must be a hepatic vein and Q a hepatic artery.

D The organ is a kidney; P must be the renal vein and Q the renal artery.

E The organ is skeletal muscle; P must be a vein and Q an artery.

WORKED SOLUTION

From P to Q, oxygen falls and carbon dioxide rises, consistent with respiration by the organ's cells.

Urea falls sharply, indicating removal from the blood. This strongly fits a kidney.

Blood enters a kidney in a renal artery and leaves through a renal vein, so P could be the renal artery and Q the renal vein.

Answer B.

B-18 Digestion · bile and lipase Answer **B**

QUESTION 18

Equal amounts of lipid and lipase are used in four experiments.

tube	lipid	bile	pH	time to fixed endpoint
A	large droplets	absent	8	120 s
B	mechanically emulsified into small droplets	absent	8	45 s
C	large droplets	present	8	50 s
D	large droplets	present	2	>300 s

Which conclusion is best supported?

A Bile is itself the enzyme that hydrolyses lipid.

B In these experiments, an important effect of bile could be increasing lipid surface area, because mechanical emulsification produces a similar increase in rate.

C Lipase has its optimum at pH 2.

D Bile converts lipid directly into glucose.

E Lipase functions only when bile is present.

WORKED SOLUTION

Tubes B and C reach the endpoint in very similar times even though B has no bile. In B the lipid has instead been mechanically broken into small droplets.

This supports the idea that an important effect of bile is emulsification, which increases the lipid surface area available to lipase.

Tube D also shows that an unsuitable low pH greatly reduces the reaction rate.

Answer B.

B-19 Homeostasis · ADH Answer E

QUESTION 19

A drug prevents ADH from increasing the water permeability of the collecting ducts but does not directly affect filtration or glucose reabsorption.

A dehydrated person takes the drug.

Compared with a healthy dehydrated person, which changes are expected?

1. A greater urine volume.
2. A lower concentration of urea in the urine is likely because more water remains in the urine.
3. Glucose must appear in the urine because ADH is absent from the collecting duct.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

ADH normally increases water permeability of the collecting ducts, increasing water reabsorption.

If that effect is blocked, less water is reabsorbed, so urine volume is greater and dissolved urea is likely to be less concentrated.

ADH is not the mechanism responsible for glucose reabsorption, so glucose does not have to appear in the urine.

Statements 1 and 2 only — answer E.

B-20

Nervous and hormonal coordination

Answer E

QUESTION 20

A person's hand touches a very hot surface.

The hand is rapidly withdrawn. A few seconds later, adrenaline contributes to an increased heart rate.

1. Withdrawal of the hand can be coordinated through a reflex pathway involving a relay neurone in the central nervous system.
2. Neurotransmitter molecules can diffuse across synapses in the reflex pathway.
3. Adrenaline travels from an endocrine gland to the heart along motor neurones.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

A rapid withdrawal reflex can involve a relay neurone in the central nervous system.

At chemical synapses, neurotransmitter molecules diffuse across the synaptic gap.

Adrenaline is a hormone and is transported in the blood, not along motor neurones.

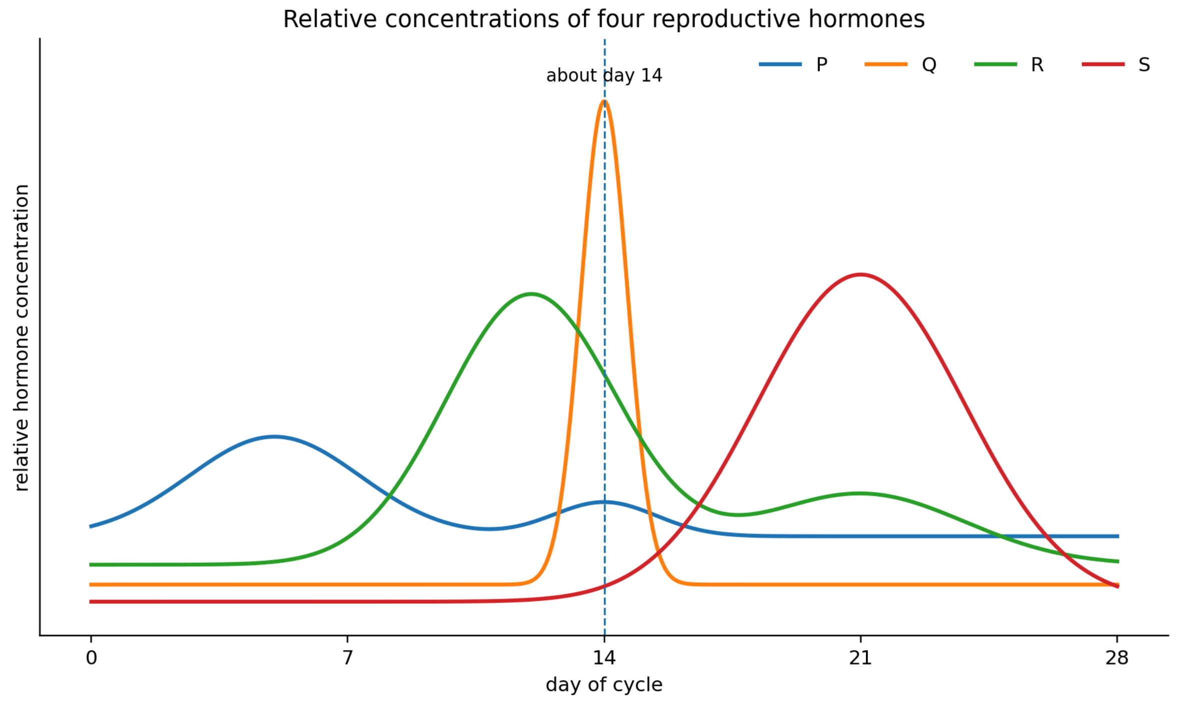
Statements 1 and 2 only — answer E.

B-21

Reproductive hormones

Answer **A**

QUESTION 21



The graph shows four hormones, P, Q, R and S.

The large narrow peak around day 14 represents the hormone that triggers ovulation. The broad peak during the second half of the cycle represents the hormone that maintains the uterine lining.

Which pair of curves represents the **two pituitary hormones whose production tends to be inhibited when oestrogen and progesterone are maintained at high levels by combined hormonal contraception**?

A P and Q

B P and R

C Q and S

D R and S

E Q and R

WORKED SOLUTION

The sharp mid-cycle peak is LH, so Q is LH.

The broad second-half peak is progesterone, so S is progesterone.

The remaining pituitary hormone is FSH, represented by P.

The two pituitary reproductive hormones are therefore P and Q.

Answer A.

B-22 Clinical trials · evidence Answer E

QUESTION 22

A medicine and a placebo are tested on two similar groups.

group	people tested	people whose symptoms improved	people reporting side effects
medicine	200	130	20
placebo	200	90	8

1. The improvement rate is 20 percentage points higher in the medicine group.
2. The proportion reporting side effects is 2.5 times as high in the medicine group as in the placebo group.
3. The trial proves that every improvement in the medicine group was caused by the medicine.

Which are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

$$\text{medicine improvement} = 130/200 = 65\%$$

$$\text{placebo improvement} = 90/200 = 45\%$$

The difference is 20 percentage points.

$$\text{side-effect ratio} = (20/200)\ /\ (8/200) = 10\%\ /\ 4\% = 2.5$$

However, improvement also occurred in the placebo group, so the trial does not show that every individual improvement in the medicine group was caused by the medicine.

Statements 1 and 2 only — answer E.

B-23

Ecology · quadrat sampling

Answer D

QUESTION 23

Ten randomly positioned **0.50 m × 0.50 m** quadrats are used to sample a plant population.

quadrat	count
1	0
2	4
3	7
4	0
5	5
6	2
7	8
8	0
9	3
10	1

The habitat has an area of **100 m^2**.

Which row gives both the estimated population and the **frequency of occurrence** of the plant?

A 300; 70%

B 600; 30%

C 1200; 30%

D 1200; 70%

E 4800; 70%

WORKED SOLUTION

Total organisms counted = 30, so the mean is 3 per quadrat.

Each quadrat has area 0.25 m^2, giving a density of:

$$3 \,/\, 0.25 = 12 \text{ plants m}^{-2}$$

$$\text{estimated population} = 12 \times 100 = 1200$$

The species occurs in 7 of the 10 quadrats, so its frequency is 70%.

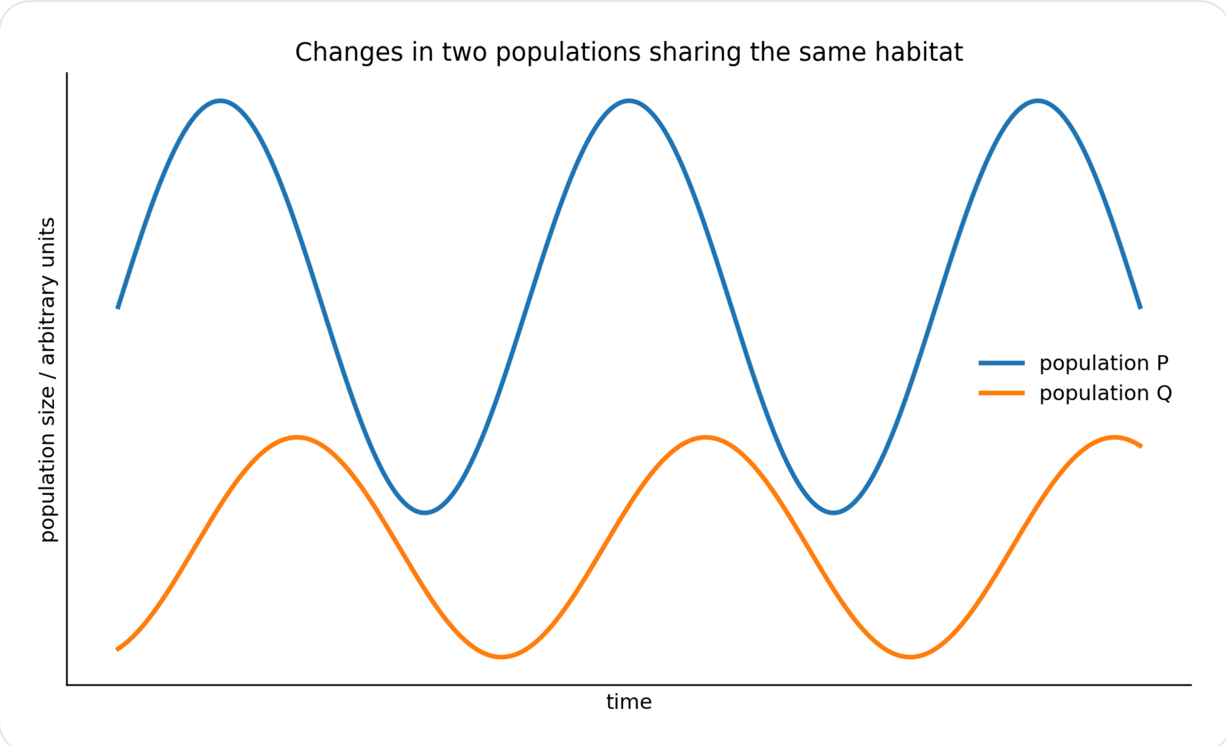
Answer D.

B-24

Ecology · population interactions

Answer F

QUESTION 24



Two populations in the same habitat show the changes in the graph.

Population Q tends to peak after population P.

1. The pattern is consistent with Q being a predator whose population partly depends on P as prey.
2. The graph alone proves that changes in P are the sole cause of changes in Q.
3. If P is an important prey species for Q, a decline in P could later contribute to a decline in Q.

Which statements are justified?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

The delayed peaks are consistent with a possible predator-prey relationship.

If P is important prey, reduced P availability could later reduce Q.

However, the graph alone cannot prove that P is the sole cause of changes in Q because other biotic or abiotic factors may also affect Q.

Statements 1 and 3 only — answer F.

B-25

Carbon cycle · decomposition

Answer E

QUESTION 25

Equal masses of leaf litter containing radioactive carbon, ^{14}C, are placed in three sealed experimental systems.

The amount of $^{14}CO_2$ released after one week is measured.

system	conditions	$^{14}CO_2$ produced / arbitrary units
A	non-sterile soil, 25°C	60
B	sterilised soil, 25°C	5
C	non-sterile soil, 5°C	20

1. Living decomposers contribute substantially to carbon release from the litter.
2. Lower temperature can reduce the rate at which carbon is returned from the litter as carbon dioxide.
3. These data show that light is essential for decomposition.

Which conclusions are supported?

- **A** none
- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

A produces far more labelled CO_2 than sterilised B, supporting a major role for living decomposers.

A and C both contain decomposers, but the lower-temperature system releases much less labelled CO_2, supporting a temperature effect.

Light was not manipulated, so statement 3 is not supported.

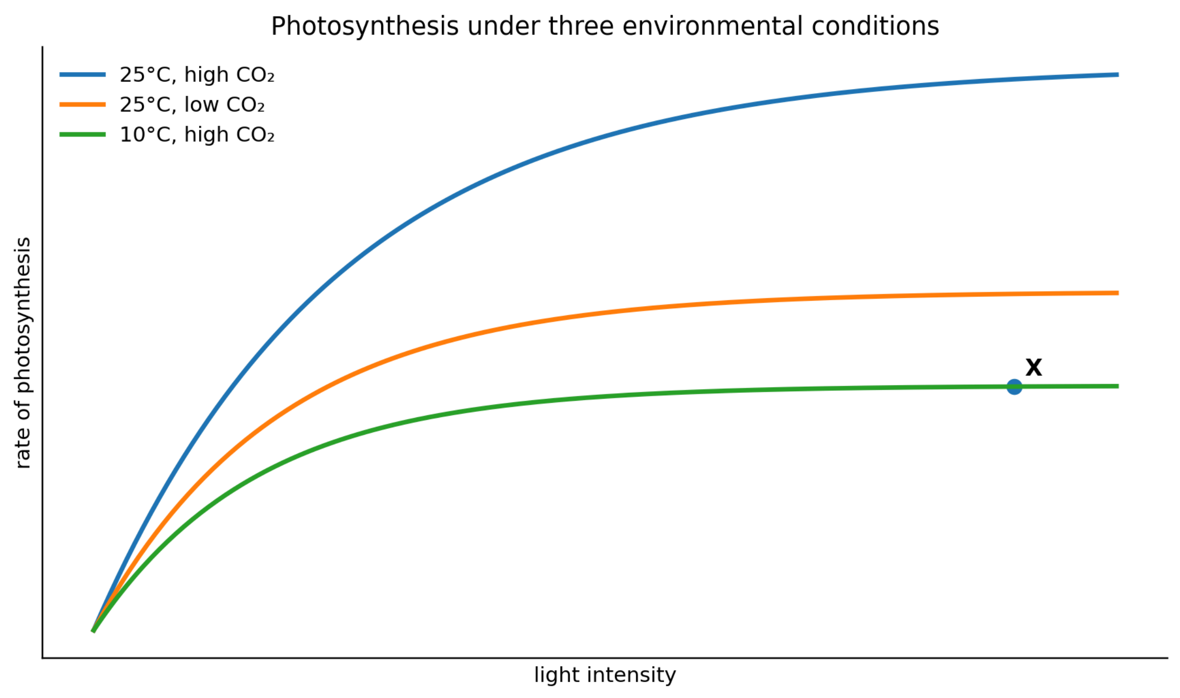
Statements 1 and 2 only — answer E.

B-26

Photosynthesis · limiting factors

Answer **C**

QUESTION 26



The graph shows photosynthesis under three conditions.

Point X is at high light intensity, high carbon dioxide concentration and **10°C**.

Which single change is most strongly supported by the graph as likely to increase the rate at X?

A Increase the light intensity further.

B Increase the carbon dioxide concentration further.

C Increase the temperature from 10°C towards 25°C.

D Decrease the carbon dioxide concentration.

E Decrease the temperature.

WORKED SOLUTION

At X, the 10°C high-CO_2 curve is already near its plateau, so increasing light further is unlikely to have much effect.

Carbon dioxide is already high.

At the same high light and high carbon dioxide concentration, the 25°C curve reaches a much higher rate. The graph therefore supports temperature as the key limiting factor at X.

Answer C.

B-27 Plant physiology · photosynthesis and translocation **Answer H**

QUESTION 27

A mature leaf is supplied with radioactive carbon dioxide, $^{14}CO_2$, for a short period.

In one plant, a ring of tissue is removed from the stem below the leaf. The treatment removes the **phloem but leaves the xylem intact**.

A third plant has an intact stem but its treated leaf is kept in darkness.

treatment	radioactivity in sugars in/above treated leaf	radioactivity in roots	relative water uptake
illuminated, intact stem	1000	400	6
illuminated, phloem removed	1300	30	6
dark leaf, intact stem	50	10	5

1. The much greater incorporation of ^{14}C in illuminated leaves is consistent with carbon dioxide being incorporated into sugars during photosynthesis.
2. The large reduction in radioactivity reaching the roots when phloem is removed supports the role of phloem in sugar transport.

3. The maintenance of water uptake after phloem removal is consistent with xylem remaining functional.

Which conclusions are supported?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

The illuminated intact leaf incorporates far more labelled carbon into sugars than the dark leaf, consistent with carbon dioxide fixation during photosynthesis.

Removing phloem reduces root radioactivity from 400 to 30 while labelled sugar accumulates in or above the treated region, supporting phloem translocation.

Water uptake remains at 6 after phloem removal, consistent with the intact xylem continuing to transport water.

All three conclusions are supported — answer H.

Compilation record: Level 2 uses Engineering Mock Test 5 for Mathematics/Physics and the corresponding Harder than ESAT Biology/Chemistry source sets. This solution book is ready to support the later construction of the matching full test.